

What must the database remember?

Start with facts about a small campus, not SQL tables.

Question	One concrete answer
Which objects?	Student S101; course DB101; section DB101 / 1
Which properties?	S101 has a name and recorded phone numbers.
Which connections?	S101 enrolls in DB101 / 1.
Which rules?	Every section belongs to exactly one course.

1. Miniworld: the part of the world represented by the database.

2. Scope: our small campus example covers Fall 2026 only.

3. Illustrative data: these are invented facts, not university policy or SQL output.

Ch1-2 introduced databases and schemas. Here, we draw a conceptual schema. No database installation is needed.

Remember: An ER model describes objects, properties, connections and rules.

What question must the design answer?

Requirements include stored facts and the operations users need.

Design work	Campus example
Collect requirements	Each section has exactly one instructor.
Check a required operation	Find every section taught by I1.
Draw the conceptual schema	Connect INSTRUCTOR and SECTION through TEACHES.
Later: logical design / mapping	Choose relational tables and their keys.
Later: physical design	Choose storage structures and indexes.

1. If the drawing cannot answer the required question, revise it.
2. If the description omits a minimum or maximum, ask for that rule; do not invent it.

Remember: ER design checks meaning before DBMS tables and storage choices.

One student is not the student type

Look at one object, then the description shared by all objects.

STUDENT describes	Properties
An entity type	StudentId, Name, Phone

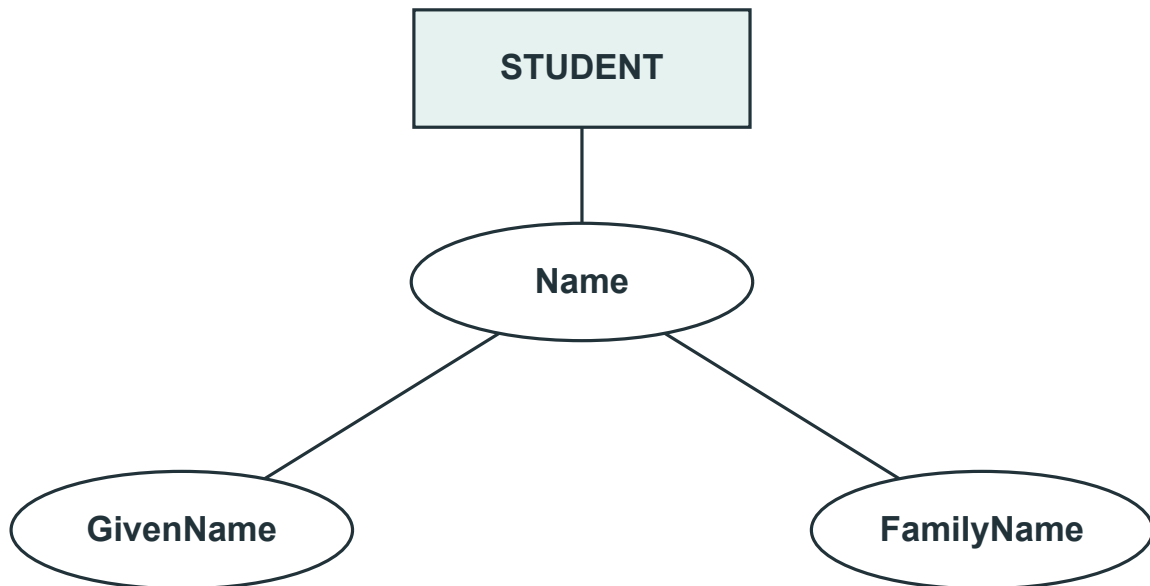
StudentId	Name	Phone values
S101	Alex Lin	02-0000-0101 02-0000-0102
S102	Blair Wu	02-0000-0201
S103	Casey Chen	None recorded

1. **Entity:** one student, identified here by S101.
2. **Entity type:** STUDENT and its shared properties.
3. **Entity set:** the students currently present. A person is not the identifier string.

Remember: Adding S104 changes the current set, not the STUDENT definition.

Split a name only when its parts matter

The application needs to search given names and family names separately.



ER diagram: an oval describes a property; connecting ovals show components.

Student	GivenName	FamilyName
S101	Alex	Lin

- 1. Name is composite:** GivenName and FamilyName are its parts.
- 2. Simple attribute:** no further components are used by this model.
- 3.** If only the whole name is needed, Name may remain simple.

Remember: A composite attribute has meaningful components in this model.

One student can have two phone values

Several values do not mean several students.



ER diagram: Phone is a multivalued attribute of STUDENT.

Student	Recorded phones	How many values?
S101	02-0000-0101; 02-0000-0102	2
S102	02-0000-0201	1
S103	None recorded	0

- 1. Single-valued:** at most one value when known.
- 2. Multivalued:** several values may belong to one entity.
- 3.** The punctuation separates invented values for reading; it is not a SQL storage design.

Remember: The double oval means multivalued, not composite.

Calculate the count from the connections

EnrollmentCount is derived; it is not another independently entered fact.



ER diagram: a dashed oval marks a derived attribute.

Student	Section enrolled in
S101	DB101 / 1
S101	CS102 / 1

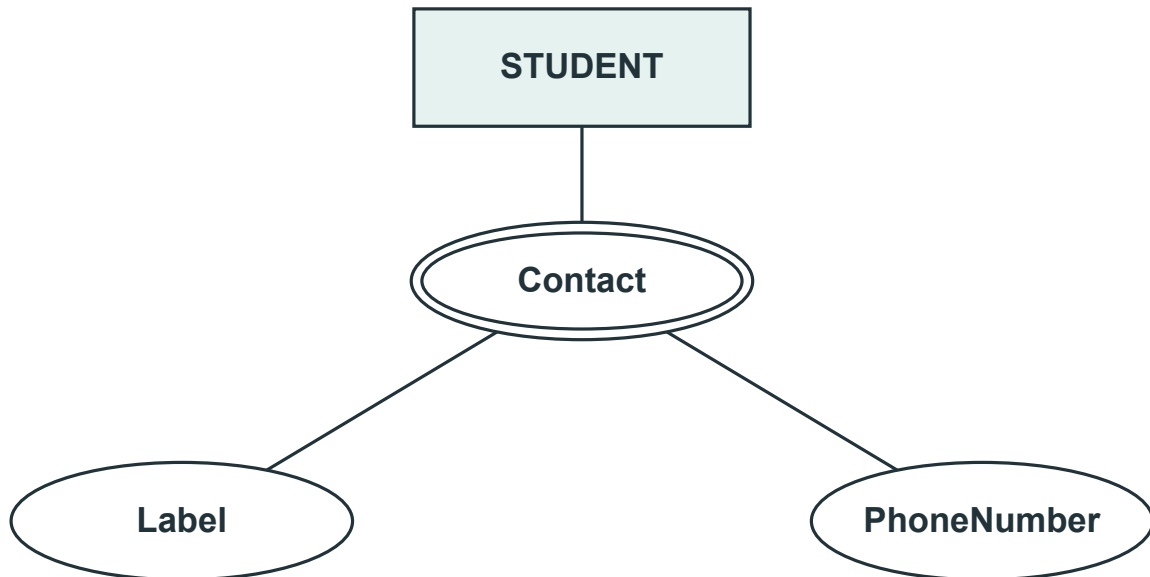
State	S101 EnrollmentCount
Both facts remain	2
Remove only CS102 / 1 enrollment	1

- 1. Stored:** information recorded directly.
- 2. Derived:** calculated from other recorded information.
3. Removing an enrollment does not change S101's identity.

Remember: Two enrollment facts give a count of 2, regardless of phone values.

A contact can have parts **AND** several values

Now each contact records a label and a phone number.



ER diagram: Contact is both composite and multivalued, a complex attribute.

Student	Label	PhoneNumber
S101	home	02-0000-0101
S101	mobile	02-0000-0102

- 1. One student; two Contact values; two components per value.**
2. Allow only one contact? Remove the double oval, but keep its parts.
3. This variant replaces the bare Phone list; it does not add a redundant second list.

Remember: Components and number of values answer different questions.

A key is a rule, not a lucky sample

Names may be different today and still repeat tomorrow.

Proposed attribute	Declared rule	A key?
StudentId	No two students share it.	Yes
Name	Different students may share it.	No



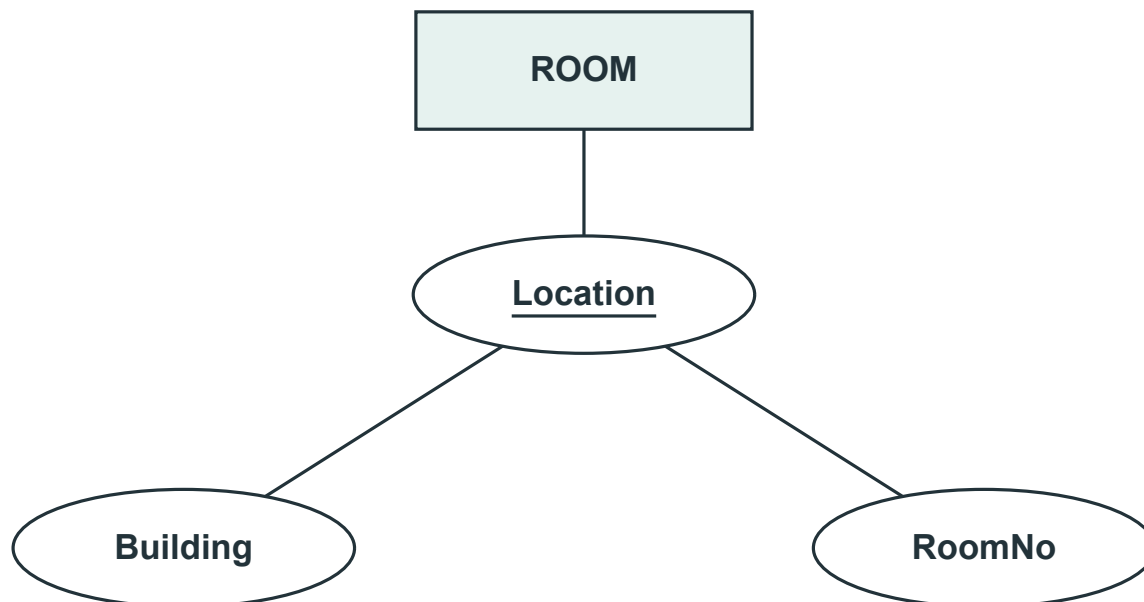
ER diagram: the solid underline marks StudentId as a key of STUDENT.

- 1. Solid underline:** each separately underlined attribute is an independent key.
- 2.** If a second attribute were also guaranteed unique, it would be another independent key, not part of one larger key.
- 3.** Choosing a relational primary key comes later, in mapping.

Remember: A key must identify an entity in every legal state.

Room 101: which building?

Use three rooms: A/101, A/102 and B/101.



ER diagram: underline Location, the composite key, not each component separately.

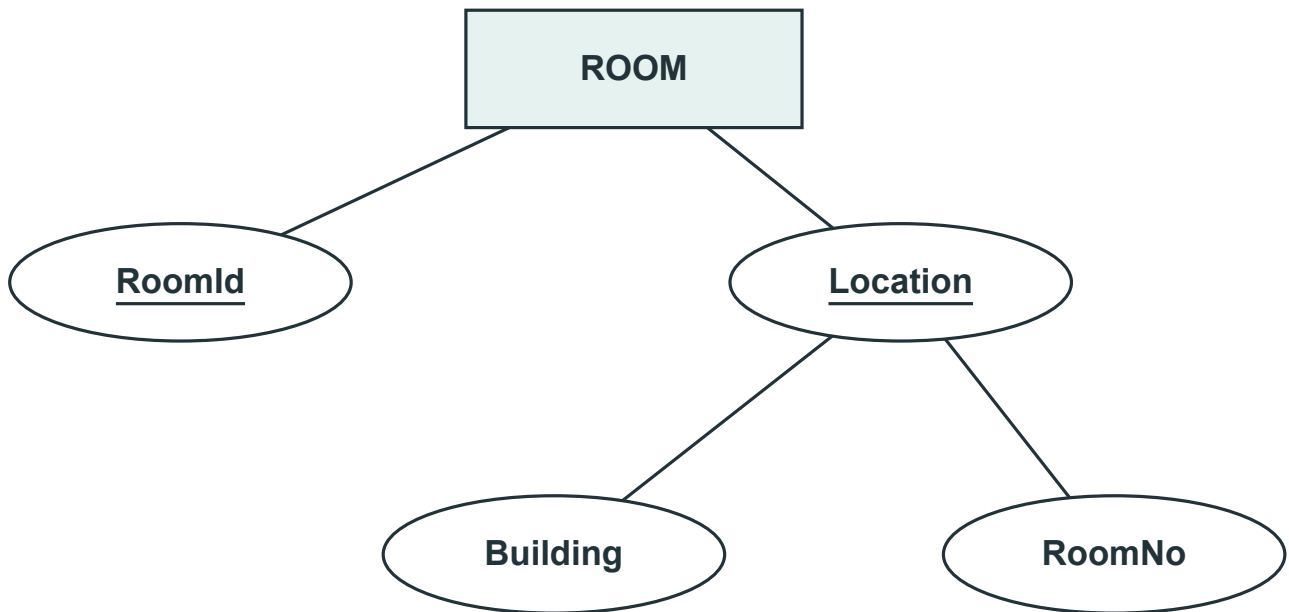
Information supplied	Matching rooms
RoomNo = 101	A/101 and B/101
Building = A	A/101 and A/102
A and 101	A/101 only

Declared rule: a room number is unique within a building. A minimal key needs both components; extra unnecessary attributes do not belong in it.

Remember: Building and RoomNo are needed together; neither identifies a room alone.

Two independent keys are not one bigger key

Keep the Location rule and add a RoomId unique across all rooms.



Two solid underlines mark two independent keys. Components are not independently underlined.

RoomId	Building	RoomNo
R1	A	101
R2	A	102
R3	B	101

1. R1 identifies one room without Building or RoomNo.
2. A plus 101 identifies that same room without RoomId.
3. Underlining Building and RoomNo separately would claim each works alone. Both repeat here.

Remember: RoomId alone works; Location works through both of its components.

Absent from the sample does not mean illegal

Example domain: Credits must be a whole number from 1 through 6.

Current course	Credits
DB101	3
CS102	2

Proposed value	Check against the declared domain
5	Allowed: whole number, within 1-6
2.5	Rejected: not a whole number
7	Rejected: above 6
0	Rejected: below 1

1. Five is allowed even though neither current course uses it.
2. A value set, or domain, constrains possible attribute values.
3. This is an illustrative rule, not the university's credit policy.

Remember: A domain gives allowed values, not just currently observed values.

Missing is not the same as zero

Ask why information is absent before interpreting it.

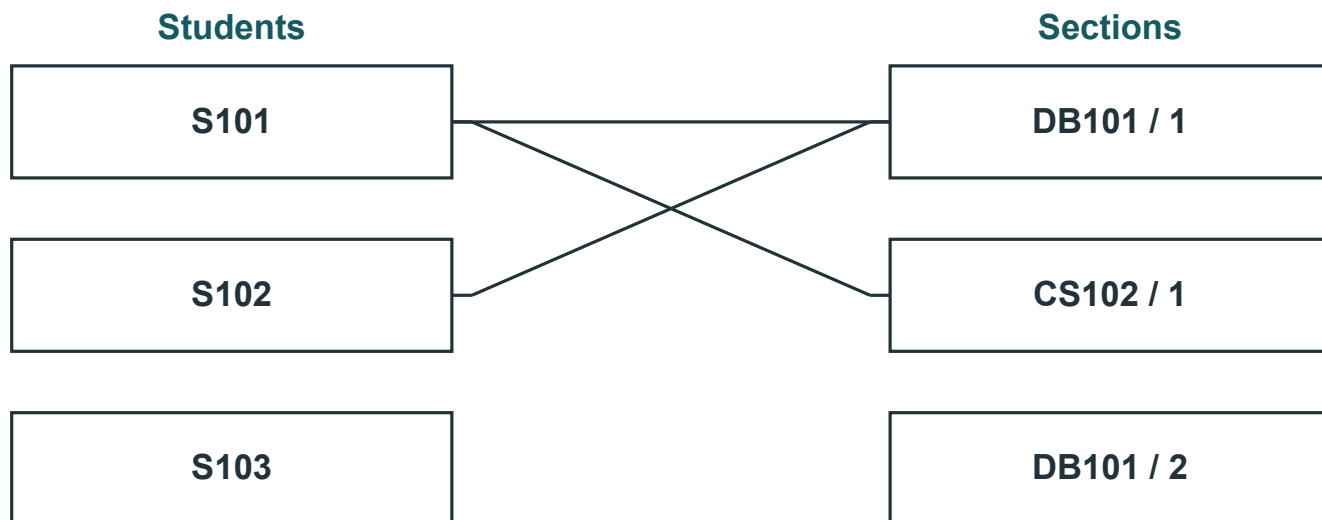
Known situation	What can we conclude?
Marking finished; recorded Grade = 0	The grade is known to be zero.
Marking is unfinished	The grade is unknown, not zero.
A person has no parking permit	A permit number is not applicable.
Nobody checked whether a permit exists	We do not know whether a number exists.

1. The textbook uses **NULL** for unknown or inapplicable information.
2. An empty entry alone does not explain which reason applies.
3. No recorded phone does not prove that the student owns no phone.

Remember: Unknown, inapplicable and a known zero have different meanings.

Count the lines, not just the students

One enrollment connects one particular student to one particular section.



Individual objects, not ER types. Each line is one enrollment; Grade is omitted here.

Inspect	Recorded participation
S101	2 lines
S103	0 lines
DB101 / 1	2 lines

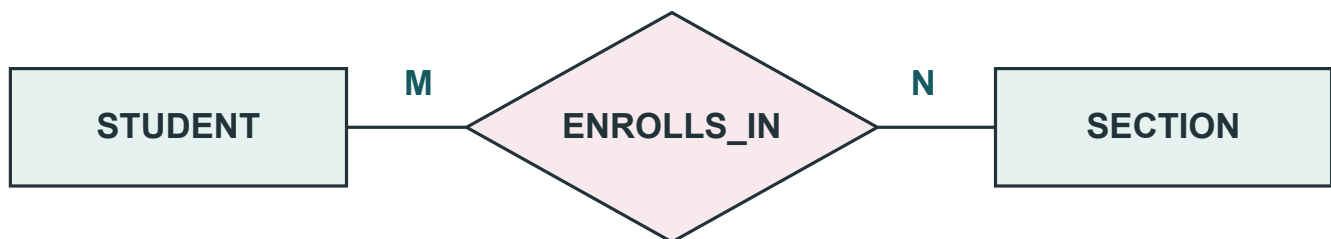
1. Relationship instance: one connection, such as S101 with DB101 / 1.

2. Relationship set: all three enrollment facts shown here.

Remember: S101 has two enrollments; S103 has none and can still exist.

The diamond describes a kind of connection

ENROLLS_IN names the relationship type, not one student's enrollment.



ER relationship close-up. Keys and SECTION identification are omitted on this page.

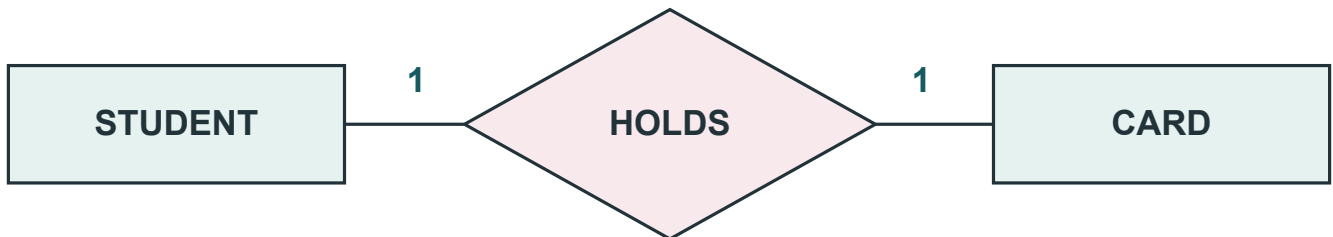
What are we counting?	Answer in the preceding example
Participating roles in each fact	2: student and section; binary
Recorded enrollment instances	3
Entity types connected by the diamond	STUDENT and SECTION

1. A rectangle names an **entity type**; a diamond names a **relationship type**.
2. Lines join participants. They are not arrows showing execution order.

Remember: Binary means two participating roles, not two recorded facts.

How many may connect on each side?

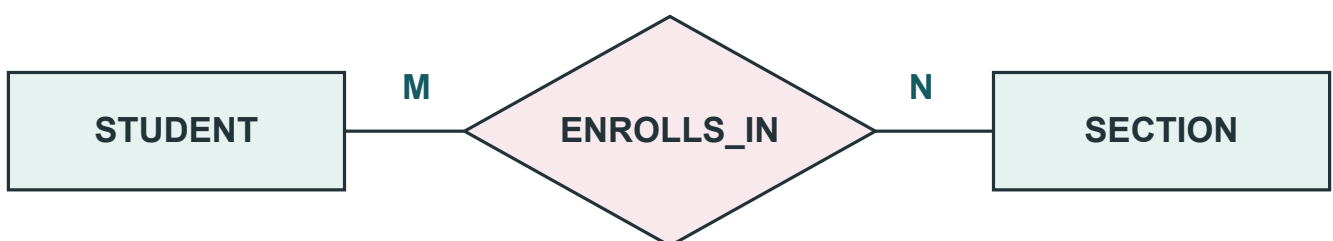
Read the maximum rule in both directions.



1:1 variant: at most one card per student; at most one student per card.



1:N: several sections per instructor; at most one instructor per section.



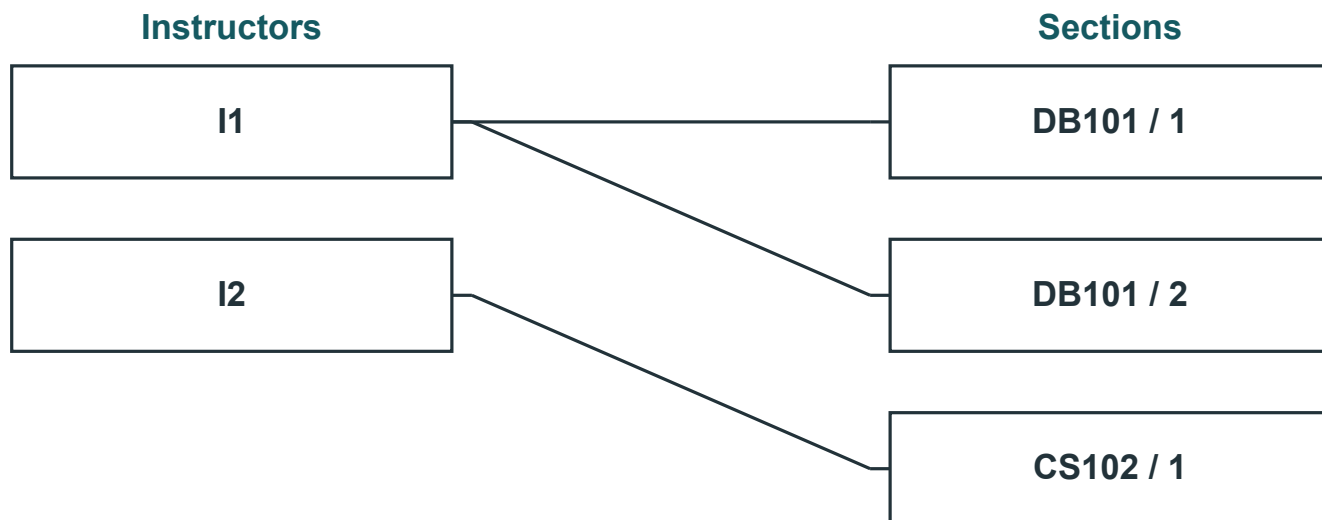
M:N: several sections per student; several students per section.

M and N specify no fixed numeric upper bound. These close-ups omit keys and identification; minima come next.

Remember: 1:1, 1:N and M:N describe permitted maxima, not today's counts.

One instructor may teach two sections

The reverse direction still permits at most one instructor per section.



Each line is one teaching assignment between individual objects.



Maximum-ratio close-up; identification symbols are omitted.

1. The **1** near **INSTRUCTOR** limits instructors for one section.
2. The **N** near **SECTION** permits several sections for one instructor.

Remember: I1 teaching twice is allowed; two instructors for DB101 / 1 are not.

Can a section have no instructor?

Now add the minimum: every section must have an instructor.



TEACHES close-up: double connecting line at SECTION; identification is omitted.

Independent proposal	Decision
Add I3 with no teaching assignment	Allowed: instructor participation is partial.
Remove DB101 / 2's only instructor	Rejected: section participation is total.
Add a second instructor to DB101 / 1	Rejected: maximum one instructor.
Replace DB101 / 2's instructor	Allowed: exactly one instructor remains.

Remember: The ratio limits the maximum; the double line requires participation.

Min-max counts an entity's participations

These are the same TEACHES rules, written in another notation.



Each pair bounds one adjacent entity's participation count. Identification is omitted.

Read beside	Meaning	Example
INSTRUCTOR (0,N)	Zero or more TEACHES instances	I1 has 2; I3 may have 0.
SECTION (1,1)	Exactly one TEACHES instance	DB101 / 1 has 1 instructor.

1. Minimum first, maximum second.

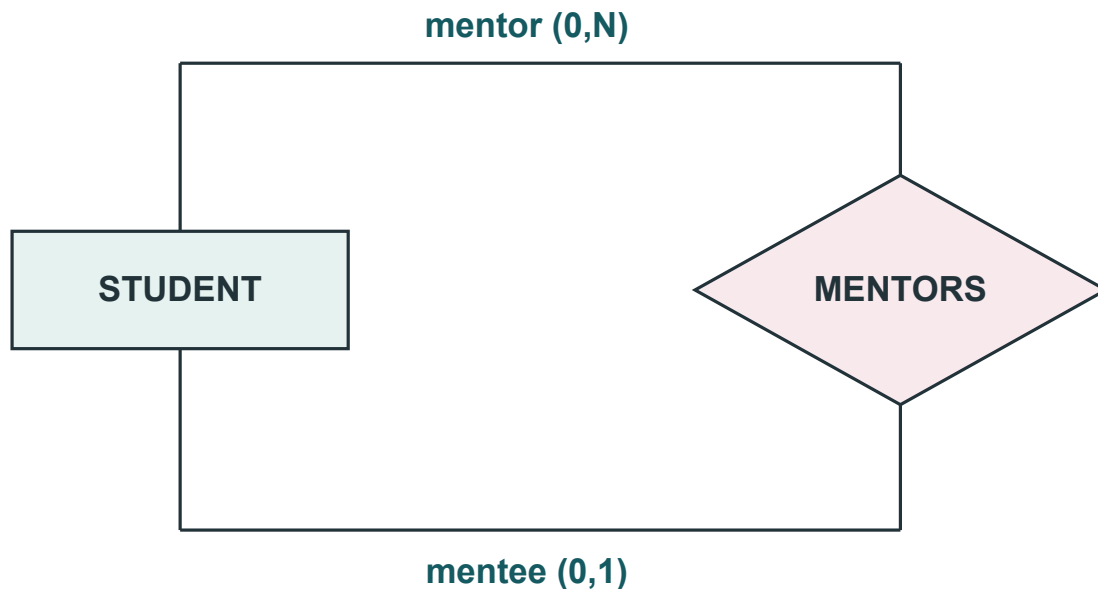
2. (1,1) beside SECTION does not mean one student per section.

3. The pair supplies both bounds; a double participation line is unnecessary here.

Remember: Do not copy label positions from maximum-ratio notation.

Same students, different roles

S101 mentoring S102 is not the same fact as S102 mentoring S101.



ER diagram: STUDENT participates once as mentor and once as mentee.

Mentor	Mentee
S101	S102
S101	S103

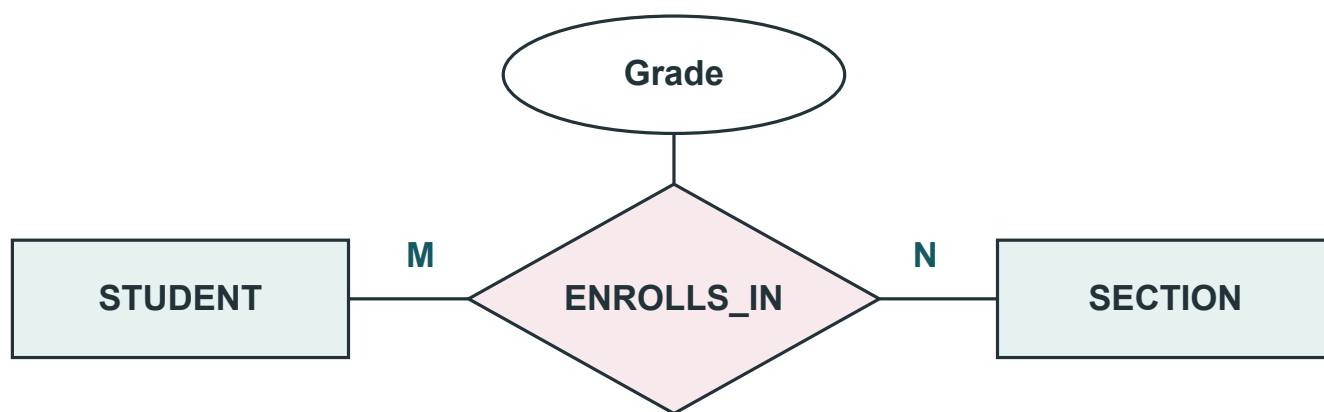
1. S101 has two participations as **mentor**, zero as mentee.
2. One mentee may have at most one mentor; one mentor may have several mentees.
3. These counts alone do not forbid self-mentoring or cycles. Those need additional rules.

Remember: A recursive relationship can be binary with just one entity type.

Whose grade is 80?

Choose both a student and a section before reading a grade.

Student	DB101 / 1	CS102 / 1
S101	80	90
S102	70	No enrollment



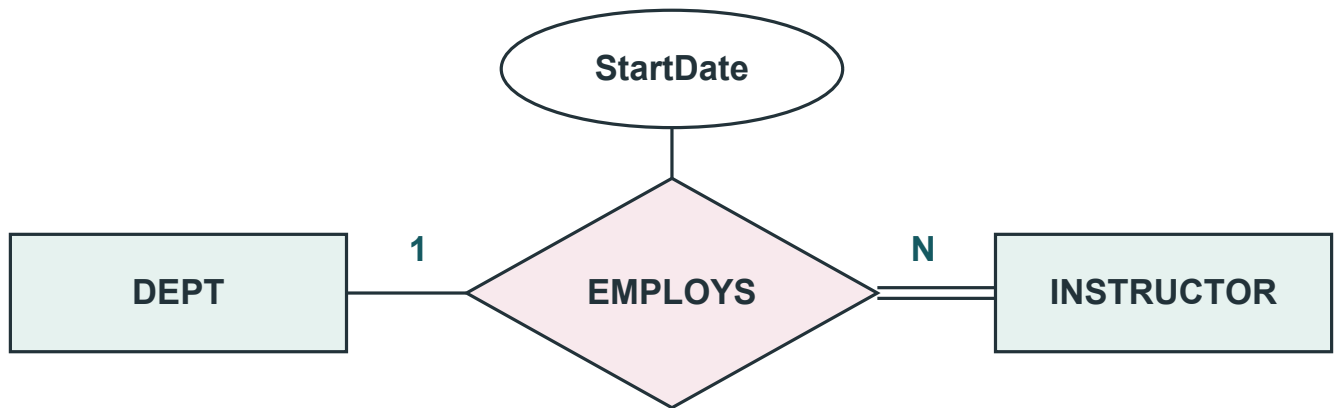
ER relationship close-up: Grade attaches to ENROLLS_IN. Identification is omitted.

1. S101 has **80 and 90**. One student-level grade cannot describe both.
2. DB101 / 1 has **80 and 70**. One section-level grade cannot describe both.
3. No enrollment, an enrollment with an unknown grade, and a recorded zero are different.

Remember: Grade belongs to this M:N enrollment, not to **STUDENT** or **SECTION** alone.

Can a relationship attribute move to an entity?

Sometimes. First check how many relationship instances one entity can have.



Current employment only: each instructor has exactly one department. Keys omitted.

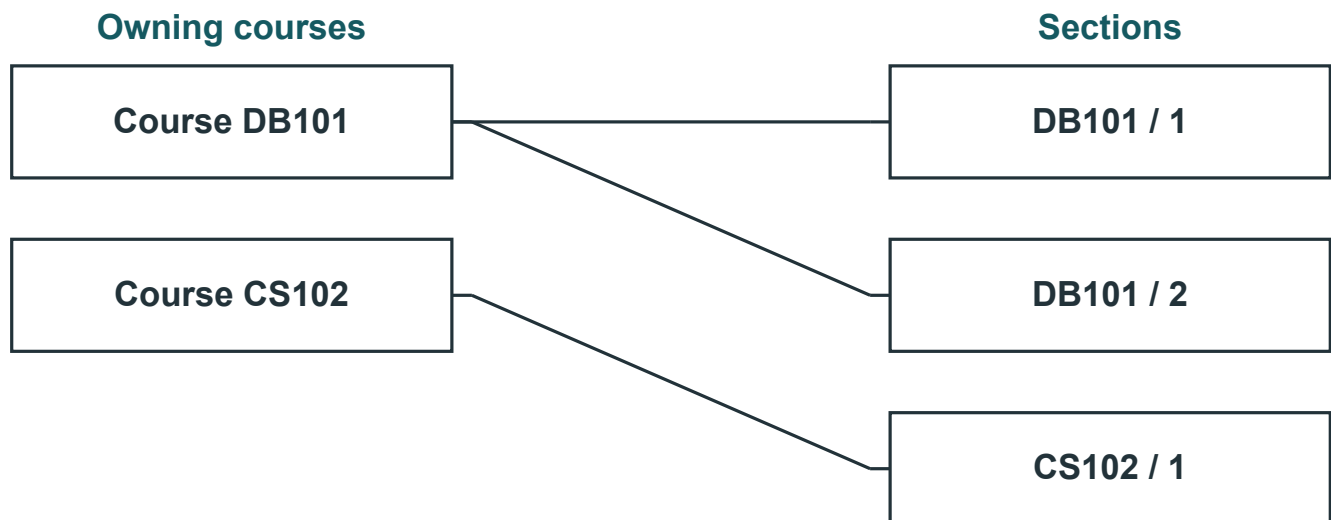
Relationship pattern	When entity placement preserves the association
1:N current employment	StartDate may go on INSTRUCTOR, the N-side: one current employer.
1:1 current chair appointment	Its date may go on either participant: each appears at most once.
M:N enrollment grades	A student's different grades need the student-section combination.

Employment history would change the problem. Several past appointments need their separate dates and identities.

Remember: Placement depends on the relationship's constraints, not one universal rule.

Section 1: which course?

Different courses are allowed to reuse a section number.



Individual classes in Fall 2026 only. The slash separates owner and local number.

- 1. Local rule:** no repeated SectionNo within one course.
- 2. No independent ID:** SECTION has no global SectionId or other complete key of its own.
- 3.** Point to the two section-1 objects. Their owners differ.

Remember: DB101 / 1 and CS102 / 1 are different classes, not duplicate records.

Find the course, then the section number

Keep the same three classes. Count what each description can identify.

Information supplied	Matching classes	Count
SectionNo = 1	DB101 / 1; CS102 / 1	2
Course = DB101	DB101 / 1; DB101 / 2	2
Course = DB101 AND SectionNo = 1	DB101 / 1	1

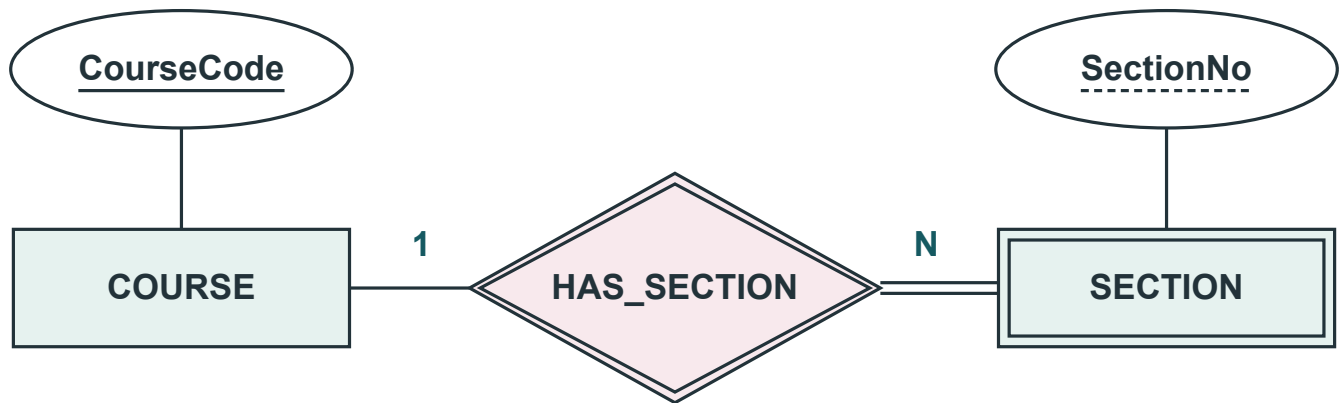
1. **Owner:** COURSE supplies the course identity.
2. **Partial key:** SectionNo distinguishes sections after their course is fixed.
3. **Weak entity type:** SECTION has no complete key of its own.

The declared numbering rule guarantees identification. Three sample rows alone cannot establish a key.

Remember: Owner DB101 plus partial key 1 identifies one section.

Now put that identification into the ER diagram

CourseCode identifies COURSE; SectionNo finishes the choice within that course.



ER schema for the same one-term classes, not a picture of individual class objects.

Point to	Read it as
Double SECTION rectangle	No complete key of its own
Dashed SectionNo underline	Partial key within the owner
Double HAS_SECTION diamond	Identifying relationship to the owner
Double connecting line at SECTION	Every section must have its owner

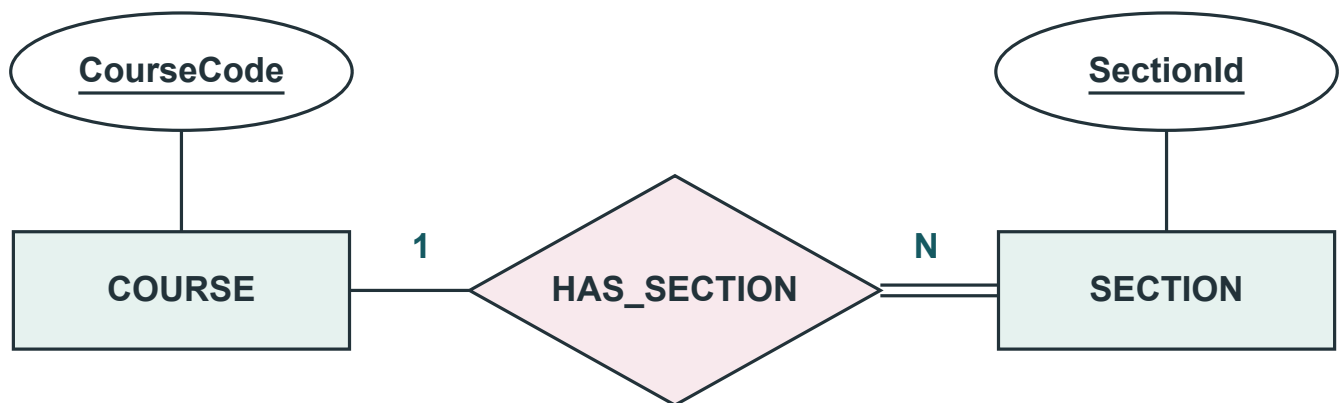
CourseCode stays on COURSE. HAS_SECTION supplies the owner; do not copy CourseCode into a SECTION oval.

Remember: Double rectangle = weak entity. Double connecting line = required participation.

A global ID changes identification

Separate variant: give the same classes their own unique SectionId values.

SectionId	Course	SectionNo
Q01	DB101	1
Q02	DB101	2
Q03	CS102	1



ER close-up: SectionNo is omitted; its local numbering rule still applies.

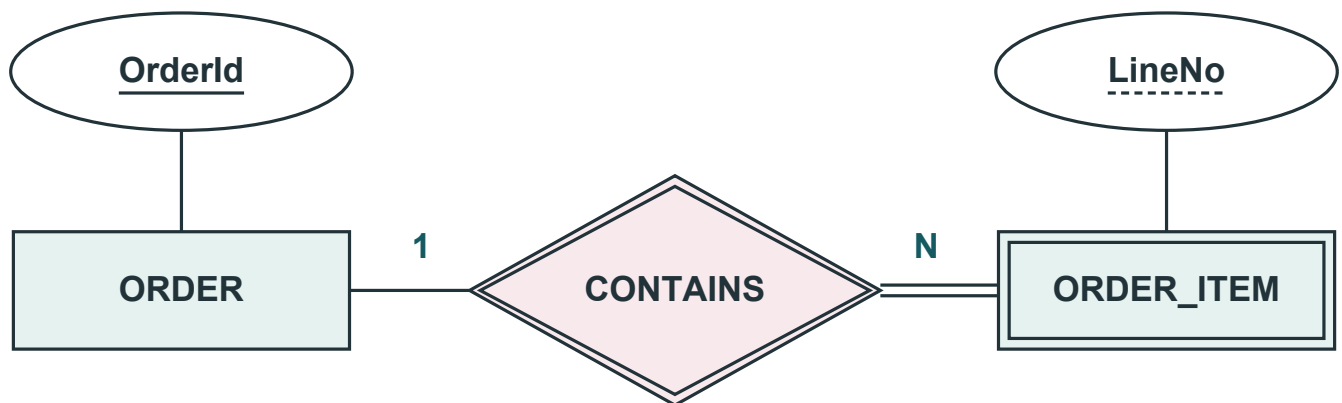
- 1. Q01 alone identifies one section:** SECTION is now regular (strong).
- The rectangle and diamond become single; SectionId has a solid underline.
- The connecting line stays double: the section must still belong to a course.

Remember: Requiring an owner does not automatically make an entity weak.

Order item 1 also needs its order

LineNo restarts in each order; there is no independent ItemId.

OrderId	LineNo	Product	Quantity
O10	1	Notebook	2
O10	2	Pen	3
O20	1	Notebook	2



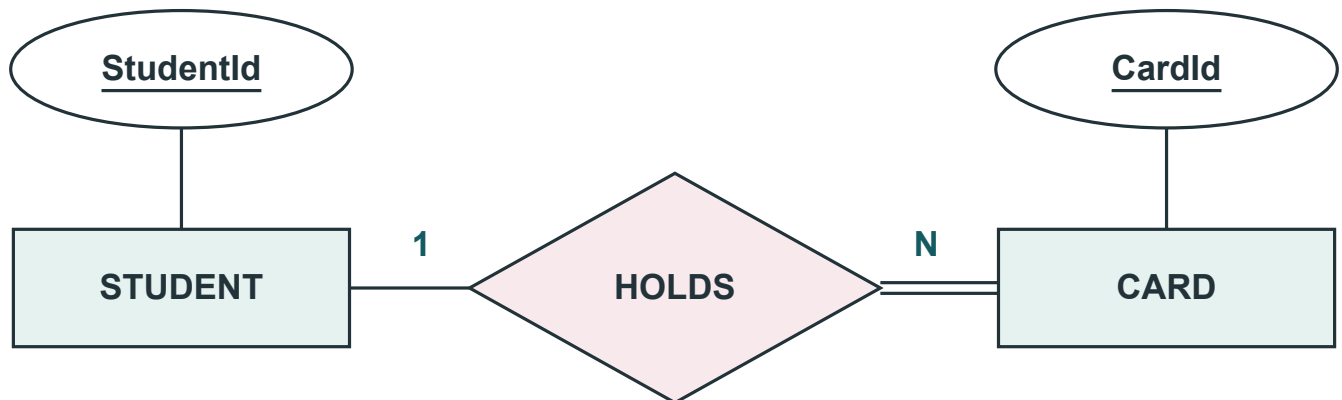
Identification close-up. Product and Quantity are descriptive ORDER_ITEM attributes, omitted here.

1. Find **ORDER O10**; then choose **LineNo 1**.
2. Product = Notebook and Quantity = 2 still match two items.
3. Changing Quantity does not create a new O10 / 1. An order may have no items while being prepared.

Remember: O10 / 1 and O20 / 1 remain different even when their descriptions match.

A card can need an owner and still be strong

Here CardId is unique across all cards; a student may have several cards.



This several-card variant differs from the earlier 1:1 example.

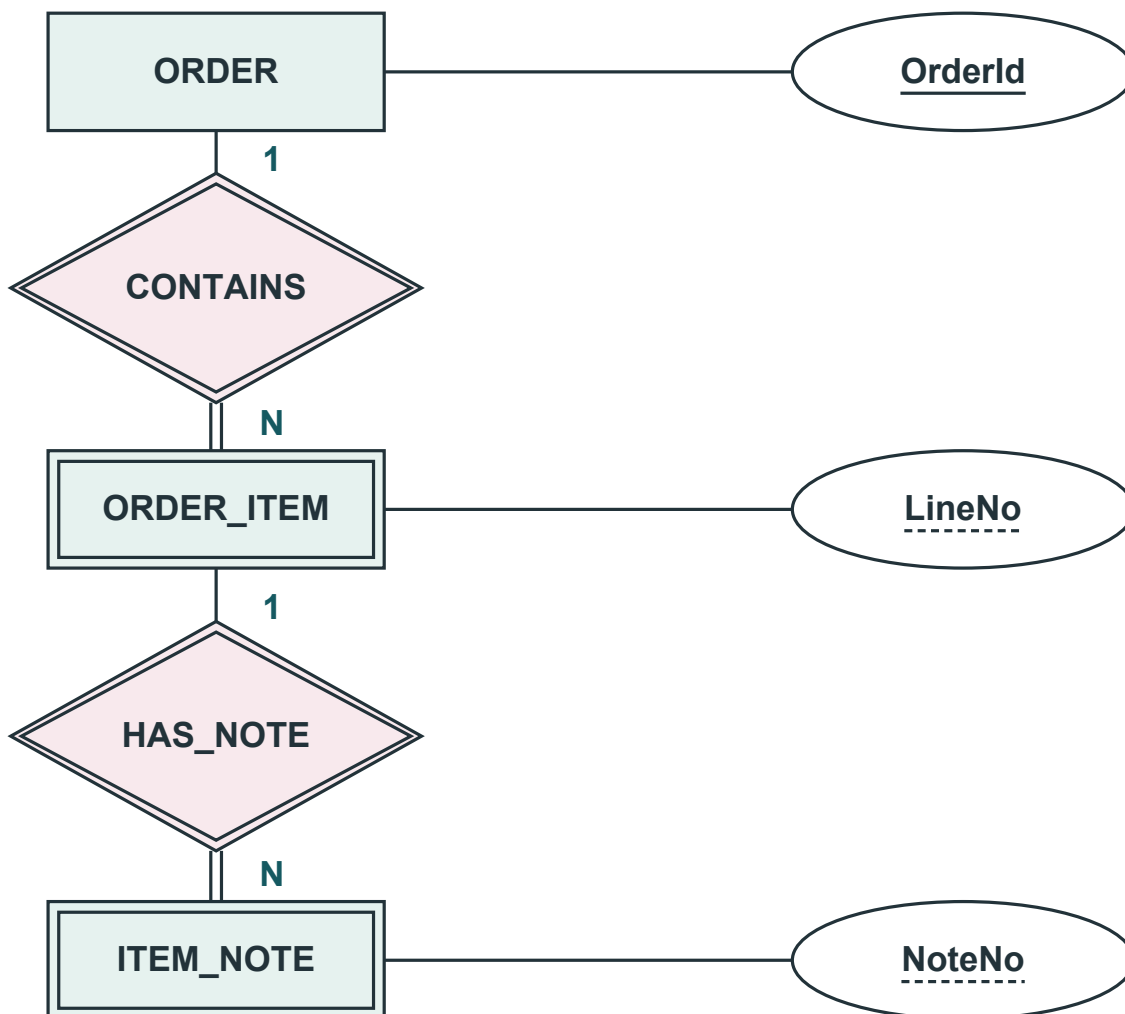
CardId	Required student owner
K10	S101
K11	S101
K12	S102

- 1. Single CARD rectangle:** its own complete key exists.
- 2. Double connecting line:** every card still needs one student.
- 3. Remove the global ID and number cards only within students:** identification would then need the owner.

Remember: K10 identifies the card without using S101's identity.

An owner can itself be weak

Add notes to an order item. NoteNo is unique only within that item.



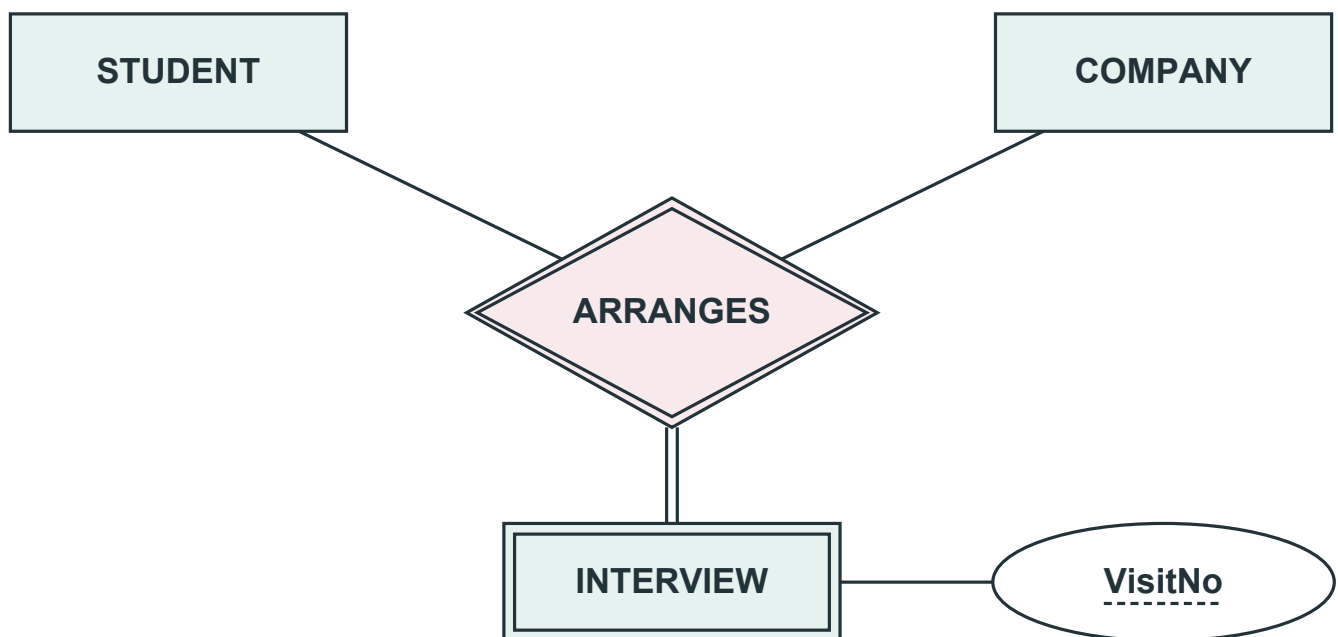
ER identification only. Product, Quantity and note text are omitted.

1. **O10 / 1 / 1**: order O10, item 1, note 1: Gift wrap.
2. **O20 / 1 / 1**: order O20, item 1, note 1: No wrap.
3. LineNo 1 plus NoteNo 1 is not enough; the order is still needed.

Remember: The note identity includes every level: OrderId, LineNo and NoteNo.

Sometimes two owners are needed together

VisitNo numbers interviews within one student-company pair.



STUDENT has key StudentId; COMPANY has key CompanyId. Owner-key ovals are omitted.

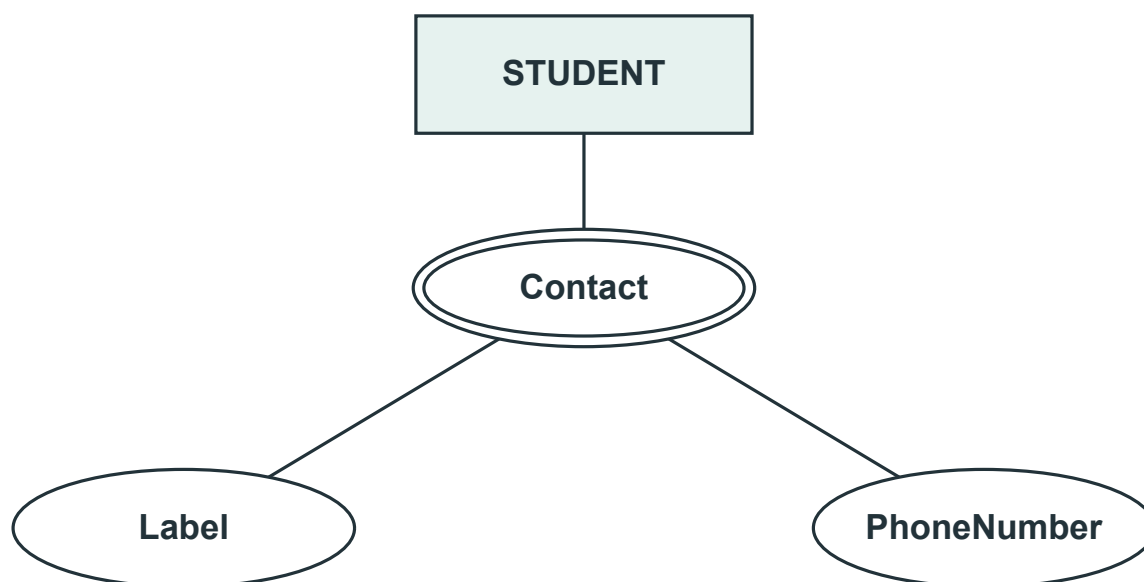
StudentId	CompanyId	VisitNo
S101	C1	1
S101	C2	1
S102	C1	1

1. Neither (S101, 1) nor (C1, 1) identifies just one row.
2. If only one interview is allowed per pair instead, the owners alone suffice; no partial key is needed.

Remember: Identify an interview by StudentId, CompanyId and VisitNo.

Several contact values do not force an entity

Choose the representation from the facts the application must record.



Attribute option: keep several (Label, PhoneNumber) values for one student.

Requirement	Possible design
Keep only labels and phone numbers	Multivalued composite Contact attribute
Record which staff member verified each contact	CONTACT entity participating in a verification relationship
Contact labels unique only within each student; no global ID	CONTACT may be weak: student identity + label

Remember: Independent relationships can justify modeling CONTACT as an entity.

Two matching names do not identify a teacher

Refine the design when instructor identity and assignments must be recorded.

Initial description	What is still unknown?
DB101 / 1: TeacherName = Morgan	Which instructor named Morgan?
DB101 / 2: TeacherName = Morgan	The same instructor, or another Morgan?

Supplied instructor fact	Supplied teaching facts
I1 has Name = Morgan	I1 teaches DB101 / 1 and DB101 / 2.



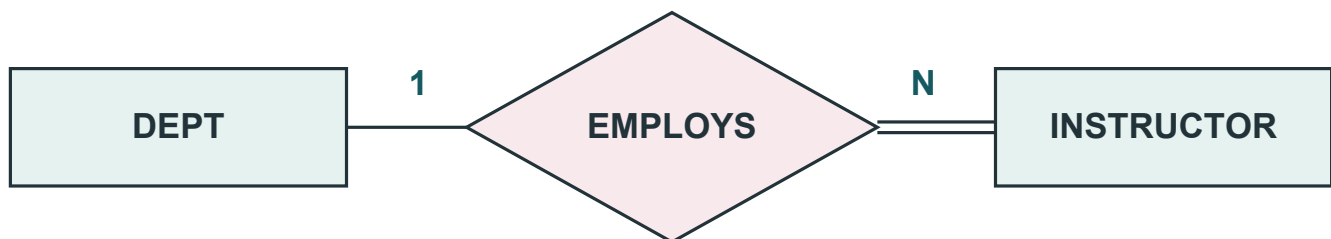
Revised relationship close-up; Name belongs to INSTRUCTOR. SECTION ownership is omitted.

1. Use singular type names, meaningful relationship names and explicit roles.
2. A noun is not automatically an entity. Ask about missing minimum and maximum rules.

Remember: Name equality is not identity; record the instructor and the assignment.

Two directions can describe the same relationship

The department of I1 and the employees of D1 must agree.



ER close-up: EMPLOYS represents both directions. Keys omitted.

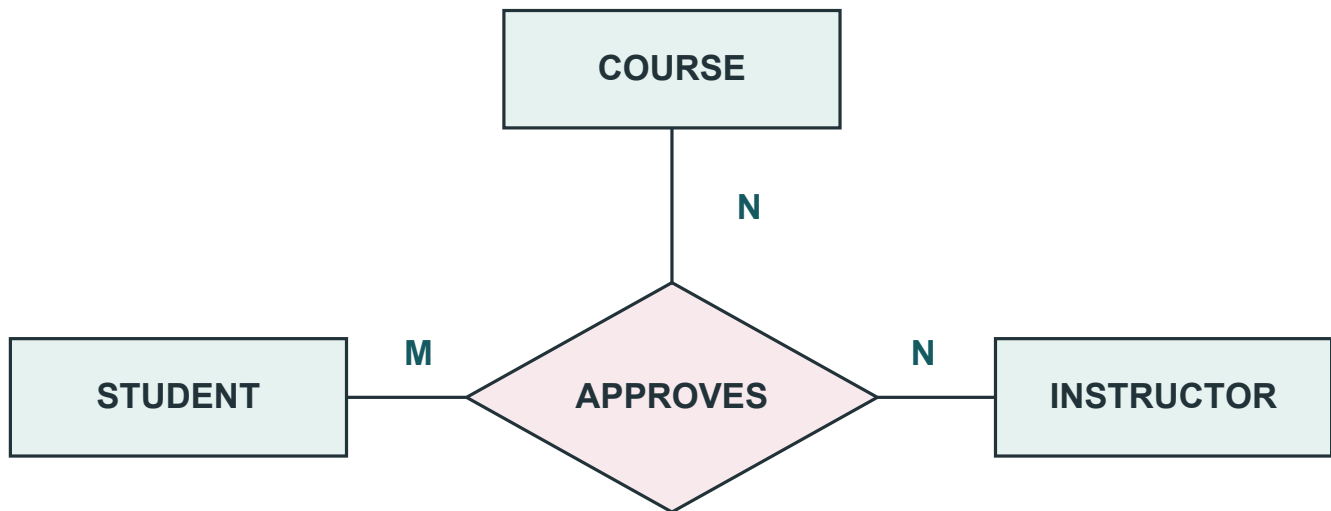
Read from an instructor	Read from the department
I1's department is D1.	D1's instructors include I1.
I2's department is D1.	D1's instructors include I2.

1. A reference-valued attribute may describe a binary relationship from one side.
2. After refining it into EMPLOYS, remove the redundant reference attribute from the ER drawing.
3. If only a department label is needed, an attribute may suffice; independent department facts justify an entity.

Remember: Use the relationship once; avoid two conflicting descriptions of the same facts.

This approval needs three participants

Who approved which student for which course? Keep the whole fact.



ER close-up: three participating roles, not a sequence of actions. Keys are omitted.

Student	Instructor	Course
S101	I1	DB101
S101	I2	CS102
S102	I1	CS102

1. These are the **complete three recorded approvals** in state A.
2. An approval is not merely a student-instructor meeting or an instructor's course knowledge.
3. Repeated approval events for the same triple at different times need an extended model.

Remember: A ternary relationship records one simultaneous three-part fact.

All three pairs exist, but the triple does not

Look for approval (S101, I1, CS102) in the preceding records.

Pair we can find	It came from
S101 + I1	S101, I1, DB101
S101 + CS102	S101, I2, CS102
I1 + CS102	S102, I1, CS102

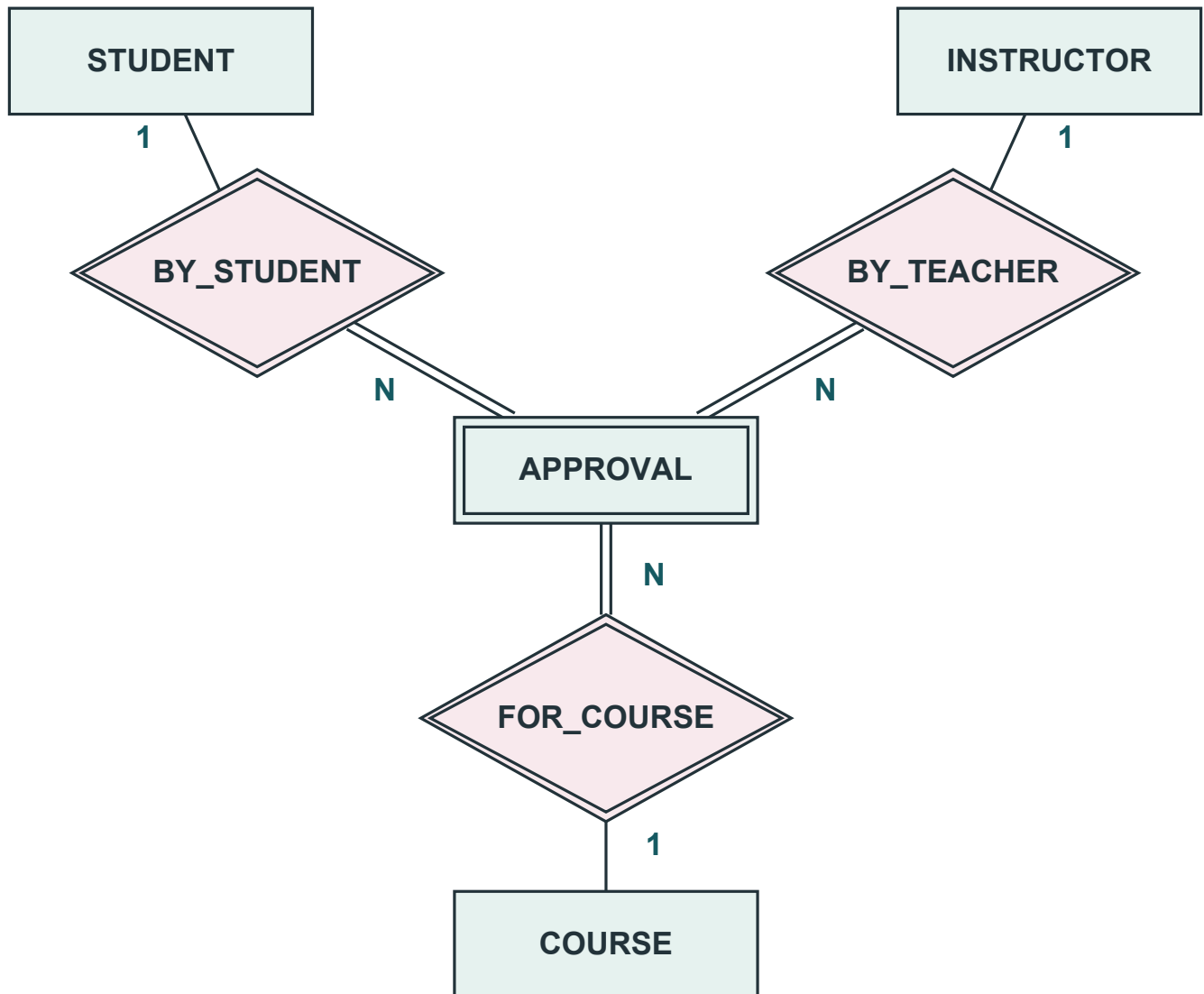
Compare	State A	State B: add the triple
Full approvals	3	4
Any new pair?	Original pairs	No
S101, I1, CS102 recorded?	No	Yes

1. The pairs come from **different records**.
2. This uses the original rules, before the later fixed-pair limit.
3. Other representations may work under additional constraints; this is not a universal ban on alternatives.

Remember: Pairwise information can lose whether a particular triple is present.

Keep the triple as one weak entity instead

Alternative representation: one APPROVAL object for each recorded triple.



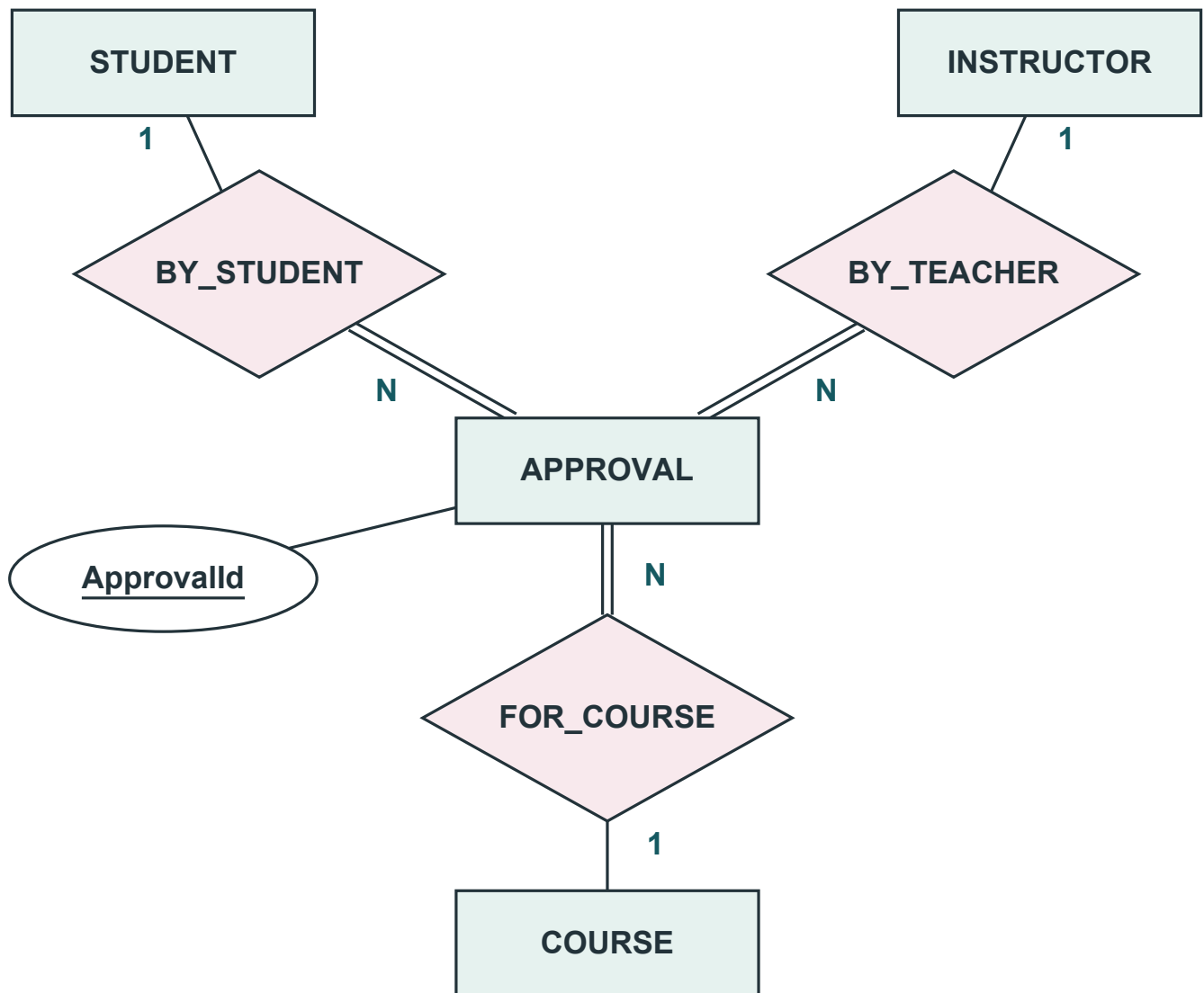
Original adaptation: StudentId, InstructorId and CourseCode are the owner keys, omitted from this view.

For S101, I1 and DB101, follow all three identifying relationships to the same APPROVAL. Do not replace it with unrelated pairwise facts.

Remember: Three owners identify APPROVAL together; no partial key is needed here.

An artificial ID makes APPROVAL regular

Give each approval an independent ApprovalId; keep its three required participants.



Regular APPROVAL: single rectangle and diamonds; required participation remains. Participant keys omitted.

A1 and A2 could have different IDs but the same three participants. ApprovalId uniqueness alone would not stop that duplicate fact.

Remember: Keep a separate uniqueness rule on the triple if one approval per triple is required.

Can extra rules make the pairs sufficient?

A restricted variant can remove the earlier ambiguity. State the extra rule.

New rule	Consequence
Each instructor is paired with only one course, and vice versa.	Instructor I1 can appear only with its paired course DB101.
Pair records are projections of the actual approvals.	S101-I1 can occur only through approval S101-I1-DB101.

Recorded approval	Recoverable with the new rule?
S101, I1, DB101	Yes: the S101-I1 pair fixes DB101.
S101, I2, CS102	Yes: the S101-I2 pair fixes CS102.

This is not the earlier general case. Eligibility or familiarity records have different meanings and must not be treated as approval projections.

Remember: Never assume the extra restriction from a small current sample.

Eligible does not mean an approval exists

New rule: an instructor may approve only courses they are eligible to approve.



ELIGIBLE_FOR is a separate binary relationship. Keys omitted.

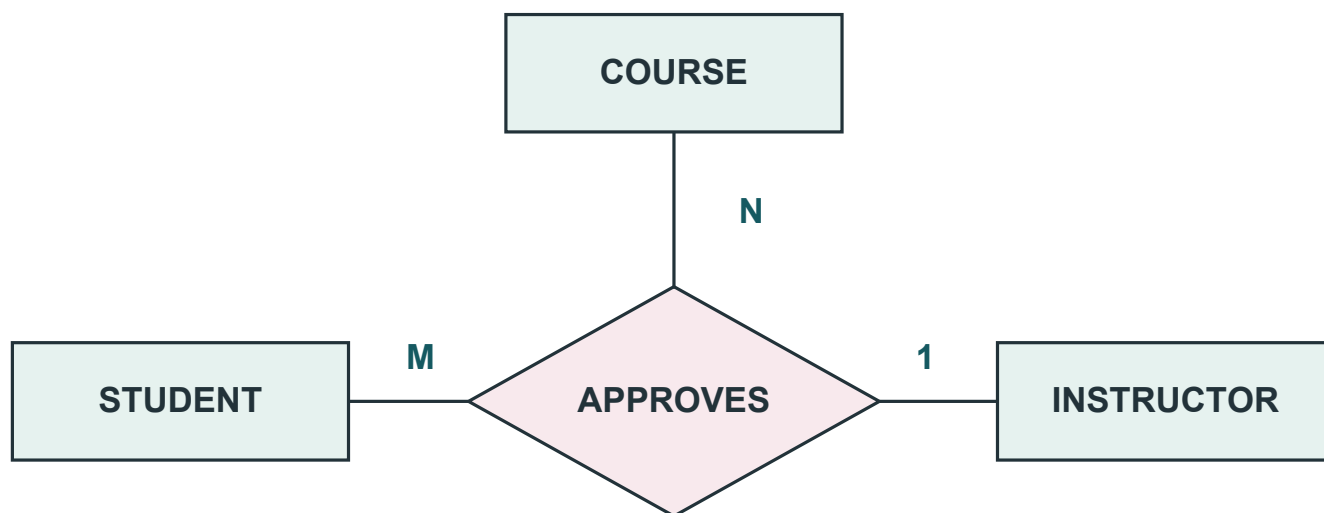
Given eligibility	Actual approval	Decision
I1 for DB101; I2 for CS102	S101, I1, DB101	Allowed
I1 for DB101; I2 for CS102	S102, I1, CS102	Reject: I1 lacks eligibility for CS102.
I1 for DB101; I2 for CS102	No approval by I2 yet	Allowed: eligibility does not require an approval.

1. Every actual approval requires the corresponding eligibility pair.
2. An eligibility pair alone does not tell us which student received approval.
3. These are new illustrative requirements, not a change to the earlier three approvals.

Remember: The link between eligibility and actual approval must be stated separately.

A ternary 1: first fix the other two

New rule: one student-course pair may have at most one approving instructor.



The 1 is beside INSTRUCTOR; first fix STUDENT and COURSE.

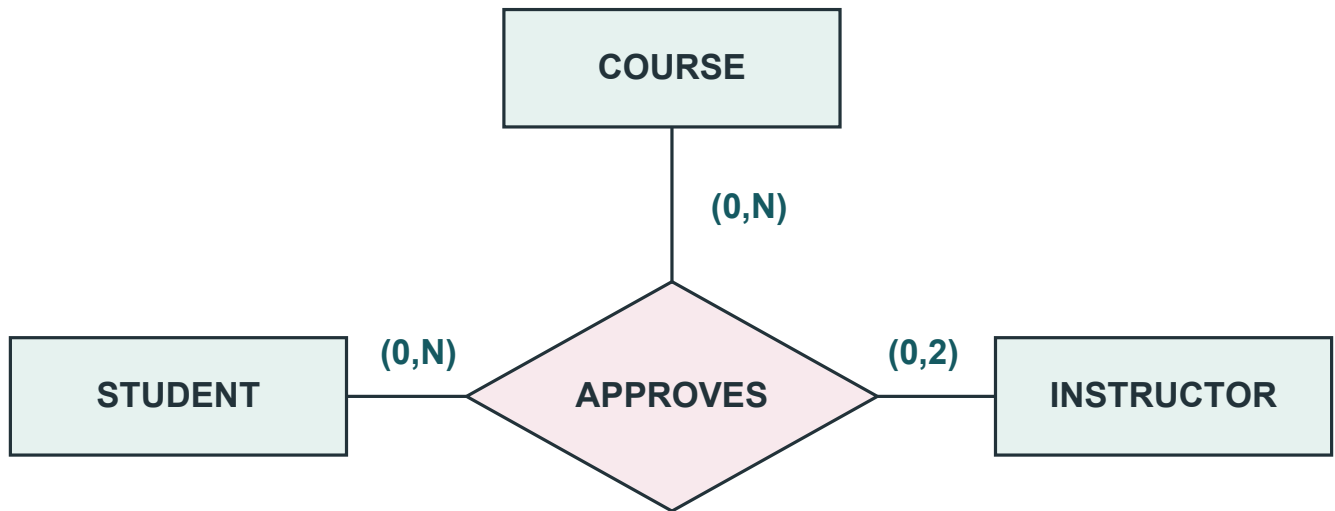
Check against the original records	Result
I1 appears with S101/DB101 and S102/CS102	Allowed: different student-course pairs.
Add S101, I2, DB101	Rejected: S101/DB101 already has I1.

1. Under this rule, the student-course pair identifies an approval instance.
2. For n roles, fix the other n-1 participants before interpreting a 1.

Remember: The 1 limits who completes a fixed pair, not the instructor's total approvals.

A ternary min-max pair counts all participations

Different rule: each instructor may appear in at most two approvals.



A separate min-max diagram. This does not replace the preceding fixed-pair rule.

State	I1 approvals	Check (0,2)
Original three approvals	2	Pass
Add S103, I1, DB101	3	Fail

1. The new student-course pair does not conflict with the preceding variant.
2. The total count still exceeds two.
3. These two kinds of ternary constraint are **not interchangeable**.

Remember: The pair (0,2) counts I1's approvals across all students and courses.

When both rules apply, run both checks

Test each proposed addition separately against the original three approvals.

Original student	Instructor	Course
S101	I1	DB101
S101	I2	CS102
S102	I1	CS102

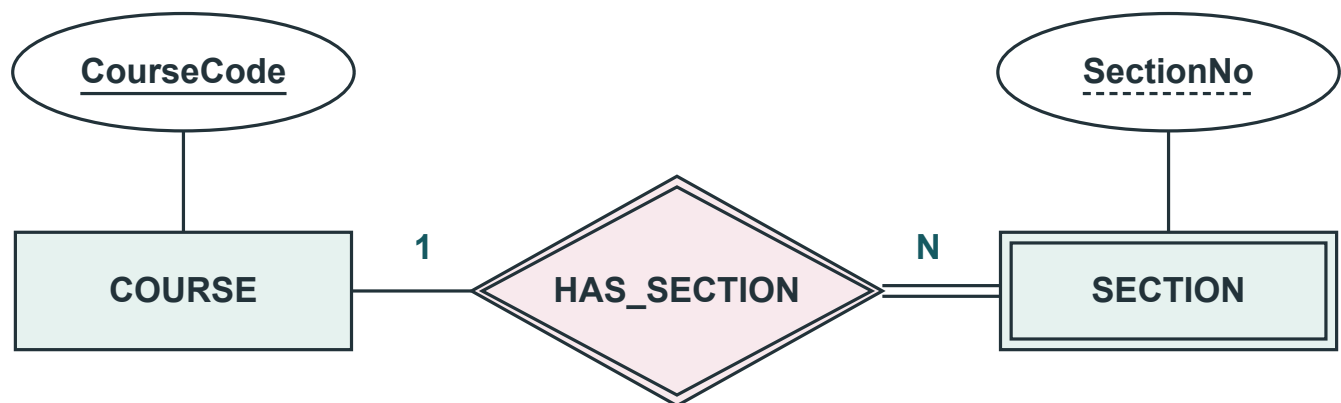
Add this approval	Fixed-pair check	Instructor total	Both?
S101, I2, DB101	Fails: I1 already approves this pair	I2: 1 to 2	No
S103, I1, DB101	Passes: new pair	I1: 2 to 3	No
S103, I2, DB101	Passes: new pair	I2: 1 to 2	Yes

Do not apply the second proposal after the first. Each row starts from the same original records.

Remember: A new pair can pass one rule while the instructor's total fails the other.

Assemble the small campus model

Use the original one-term rules, not the separate card or interview variants.



First connect COURSE and weak SECTION; keys supply identification.

Add next	Keep this rule or attribute
INSTRUCTOR TEACHES SECTION	One instructor per section; instructors may teach none.
STUDENT ENROLLS_IN SECTION	M:N, optional on both sides; Grade belongs here.
STUDENT-INSTRUCTOR-COURSE APPROVES	Keep each approval as a three-part fact.
Entity attributes	StudentId and Phone; InstructorId; CourseCode; SectionNo

Phone is multivalued. StudentId, InstructorId and CourseCode are complete keys; SectionNo is partial. Grades may be unknown.

Remember: Keep the four core relationships distinct when combining the drawing.

UNIVERSITY is a different set of requirements

Section 3.10 is not a correction to the small one-term model.

Compare	Small campus	Book's UNIVERSITY
Time span	Fall 2026 only	Several years
SECTION identity	Course + SectionNo	Global SecId
SECTION type	Weak	Regular (strong)
Students per section	Zero or more	At least five
Must have course and instructor?	Yes	Yes

1. SecId identifies a section across **all stored terms**.
2. The minimum of five is a rule in this example, not a universal database rule.
3. Do not carry the earlier weak SECTION symbols into this different design.

Remember: Changed requirements can change keys and participation constraints.

First identify the six UNIVERSITY types

Underlined keys identify objects; other attributes describe them.

Entity type	Independent keys	Other attributes
COLLEGE	CName	COffice, CPhone
DEPT	DCode; DName	DOffice, DPhone
COURSE	CCode; CoName	Level, Credits, CDesc
INSTRUCTOR	Id	IName, IOffice, IPhone, Rank
STUDENT	SId	SName, Addr, Phone, Major, DOB
SECTION	SecId	SecNo, Sem, Year, CRoom, DaysTime

1. **SName** contains FName, MName and LName.

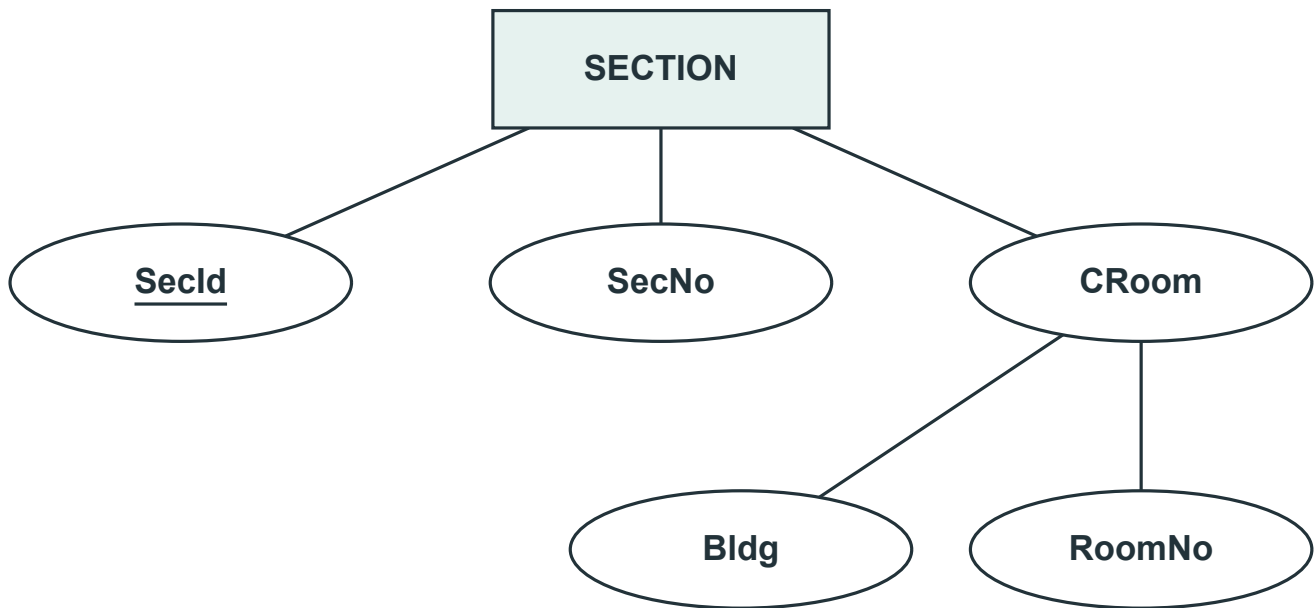
2. **CRoom** contains Bldg and RoomNo.

3. Course Level uses example codes 1-4 for undergraduate years, 5 for master's, 6 for doctoral level.

Remember: DCode and DName are separate keys; CCode and CoName are separate keys.

SecId identifies; the room components describe

SecNo can repeat in different courses or terms while SecId remains unique.



ER attribute close-up. Other SECTION attributes and relationships are omitted.

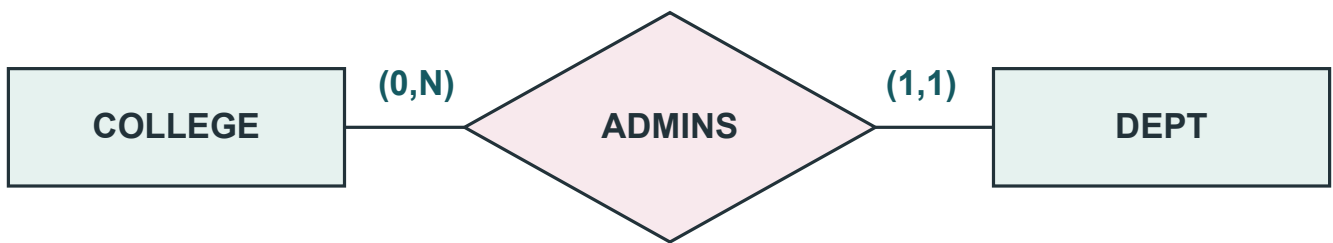
SecId	SecNo	Sem / Year	CRoom
Q101	1	Fall / 2026	A / 201
Q102	1	Spring / 2027	B / 201

1. The room numbers match, but the buildings differ.
2. Course and instructor are related entities, not extra identifying ovals copied into SECTION.

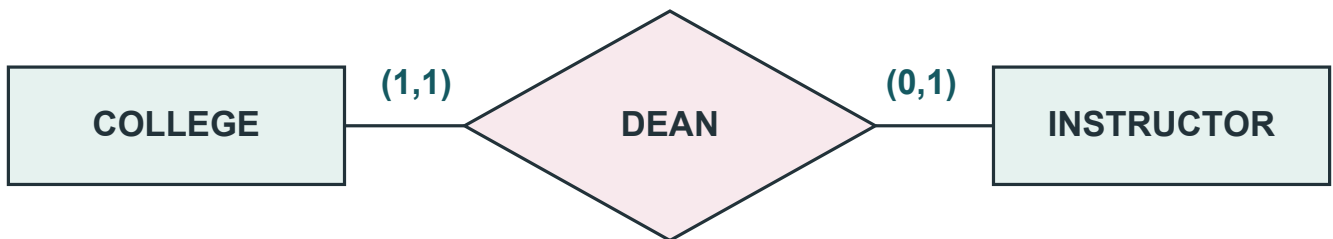
Remember: A unique SecId does not turn SecNo or a room number into a global key.

Read the organization one relationship at a time

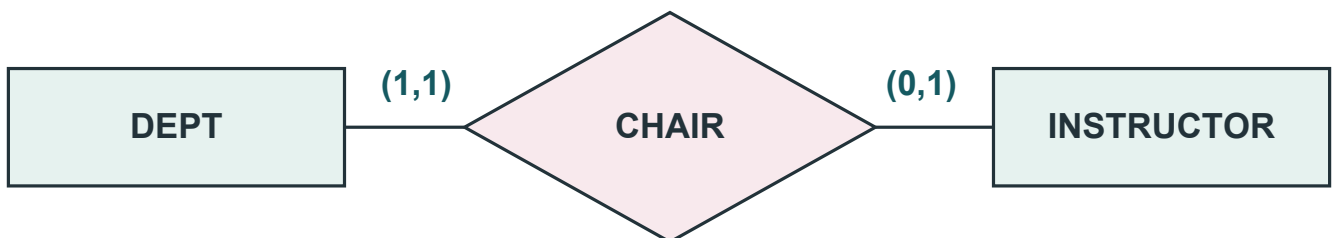
Min-max pairs bound each adjacent entity's participations. These are schema close-ups.



ADMINS: a college may administer several departments.



DEAN: an instructor may be dean of zero or one college.

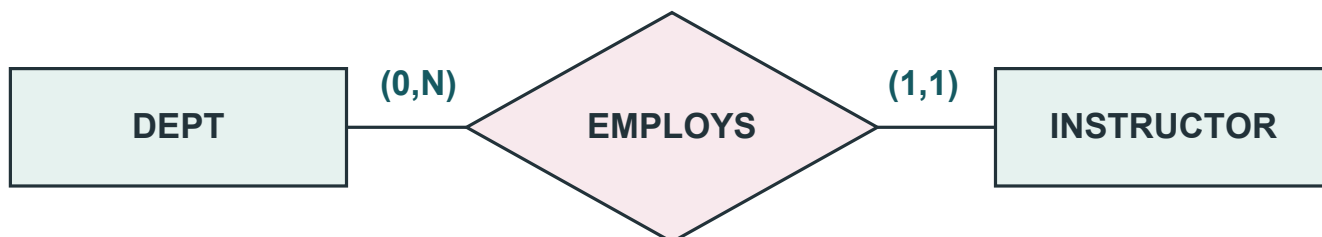


CHAIR: an instructor may chair zero or one department.

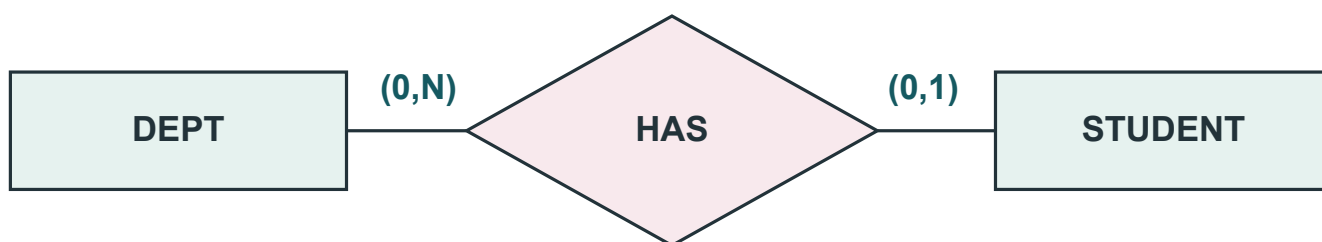
Remember: Every department has one college and one chair; every college has one dean.

Employment and a student's department differ

Read the two relationships separately, including a source discrepancy.



EMPLOYS: each instructor has exactly one employing department.



HAS as drawn in Figure 3.20: a student may have zero or one department.

Source version	Student with no department
Figure: STUDENT (0,1)	Allowed
Prose: one primary department, (1,1)	Not allowed

The photographed prose on p.123 and Figure 3.20 differ here. Do not silently combine them.

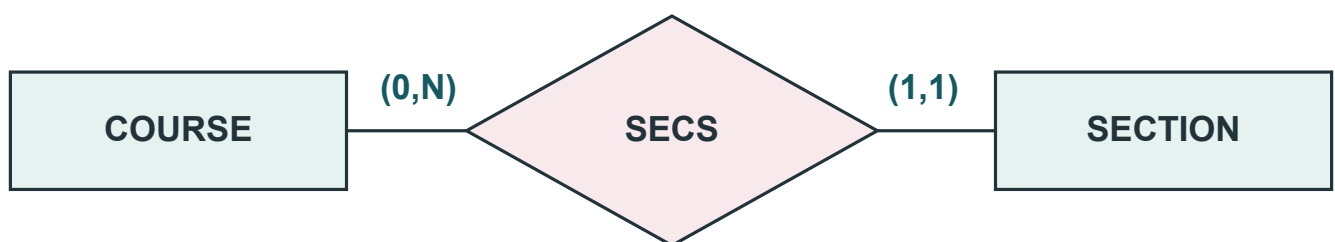
Remember: State whether you are checking the figure's HAS rule or the prose rule.

Department, course and section are not synonyms

Follow OFFERS first, then SECS.



OFFERS: each course belongs to one department.



SECS: each section belongs to one course. SECTION has its own SecId.

Invented fact	Relationship
D1 offers DB101	OFFERS
Q101 is a section of DB101	SECS

Entity keys and other attributes are omitted from these relationship close-ups.

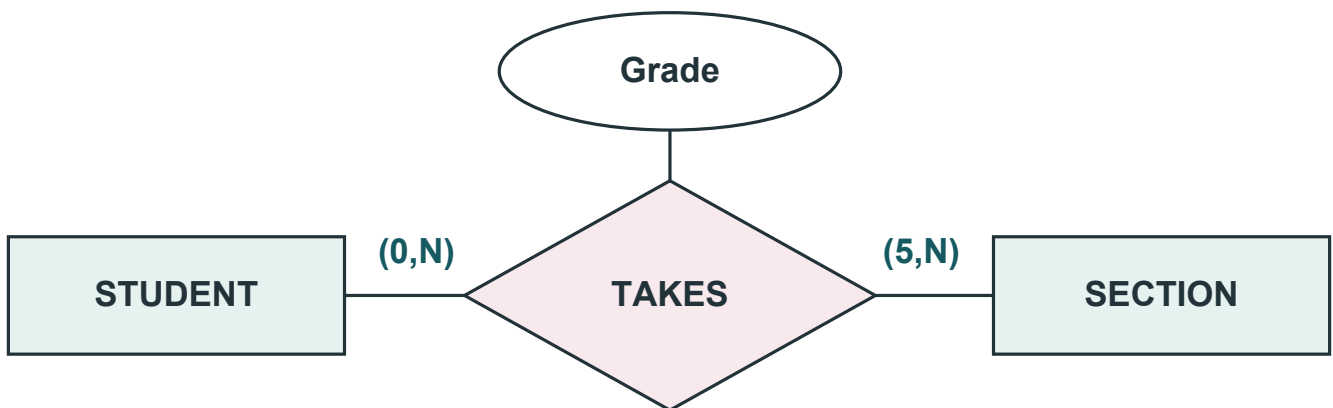
Remember: One department offers a course; one course has its sections.

A teacher and five students are different counts

Both constraints concern SECTION, but they count different relationships.



TEACHES: one instructor for each section.



TAKES: at least five students per section. Grade belongs to this relationship.

1. Q101 may be taught by **I1** while **U1-U5** take it.
2. An instructor with no teaching assignment satisfies $(0,N)$.
3. These complete all nine relationship types from Figure 3.20, shown across four pages.

Remember: Exactly one teacher does not mean exactly one student.

An employee does not have to be a chair

I3 works in D1 but chairs no department.

Relationship	Invented facts
ADMINS	C-A administers D1 and D2.
DEAN	I1 is dean of C-A.
CHAIR	I1 chairs D1; I2 chairs D2.
EMPLOYS	D1 employs I1 and I3; D2 employs I2.

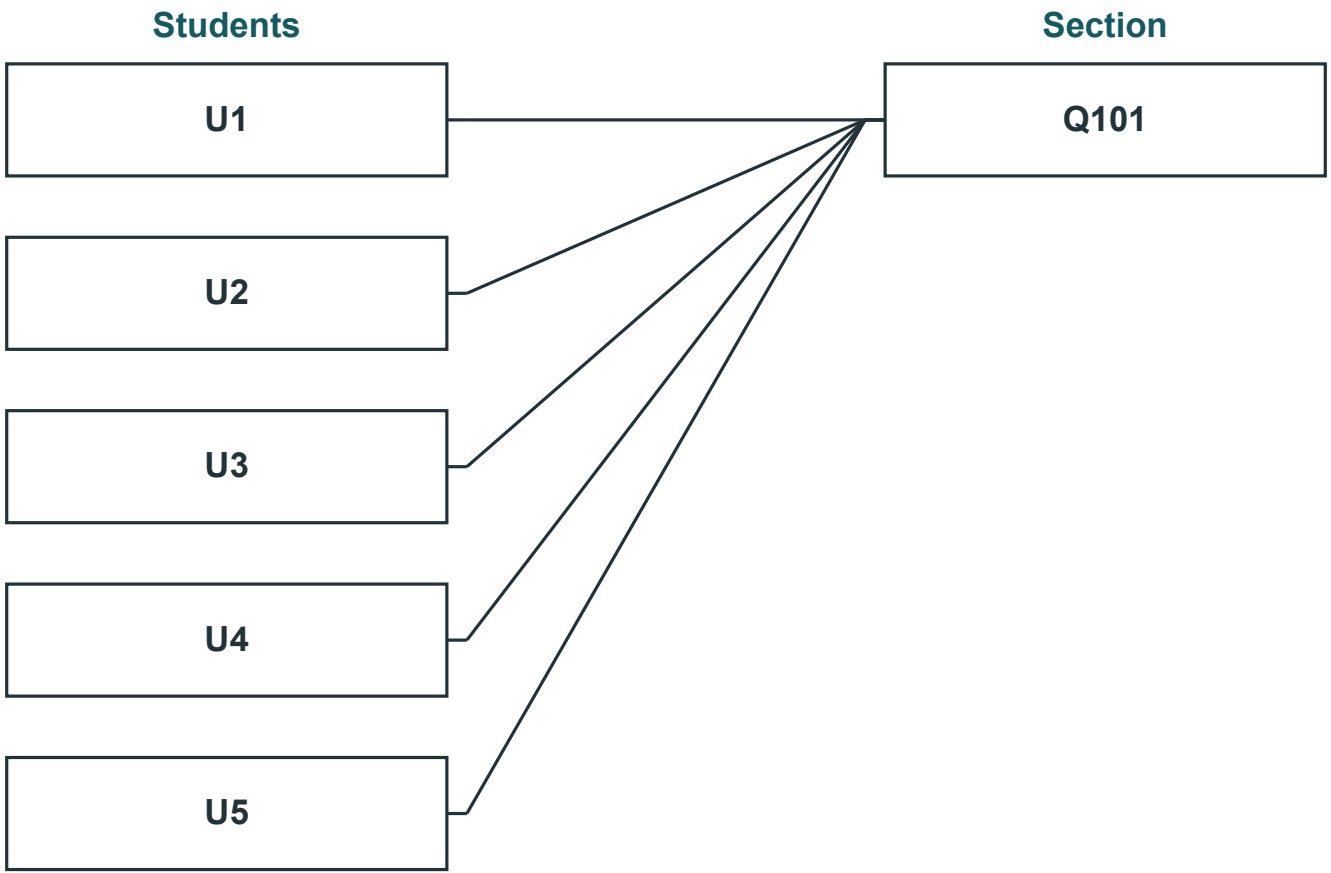
1. I3 has one employing department: **passes EMPLOYS (1,1)**.
2. I3 chairs no department: **passes CHAIR (0,1)**.
3. The sample uses chairs employed in their departments. The two relationships alone do not enforce that match.

CStartDate belongs to CHAIR: it records when an instructor began chairing that department, not when all their roles began.

Remember: Check CHAIR and EMPLOYS independently; shared membership needs another rule.

Five students become four

Keep Q101, its course and its instructor. Remove only U5's TAKES fact.



Before the change: five individual students, five distinct TAKES facts.

State	Count	Minimum 5?
Before U5 leaves Q101	5	Pass
After removing U5/Q101	4	Fail
U5 remains in STUDENT	Still 4	Still fails

Remember: Entity existence is not participation in a relationship.

Different IDs do not prevent other conflicts

Each row is a separate proposed pair of sections with different SecId values.

IDs	Shared facts	Violated rule
Q101, Q102	DB101; Fall 2026; SecNo 1	Unique course-term-number
Q201, Q202	Fall 2026; A/201; Thu-P5	No identical room-time booking
Q301, Q302	Fall 2026; I1; Thu-P5	No identical instructor-time booking

1. The instructor-time rule assumes combined sections are not permitted.
2. Changing the conflicting number, room/time or instructor may repair that rule; still check the others.
3. Different time labels may overlap. Equality alone is not a complete scheduling check.

Remember: SecId uniqueness does not enforce every numbering or scheduling rule.

From a description to a borrowing model

Original worked case: a library records individual borrowing events.

Rule	What must be represented
R1	Each BOOK represents an edition identified by ISBN. It may have no copies or many copies.
R2	Every COPY belongs to one BOOK. CopyNo is unique only within that book; there is no global copy ID.
R3	Each MEMBER has a unique MemberId and may make no loans or many loans.
R4	Each LOAN has a unique LoanId and one BorrowedOn date. It belongs to exactly one member and concerns exactly one copy.
R5	A copy may appear in no loans or many loans over time. A member may borrow that same copy again.

Draw these required attributes and relationships. Loan limits, fines and returns are outside this example.

Remember: Mark objects, identifiers and rules before choosing symbols.

Decide which identity needs an owner

Apply R1-R5 without adding a convenient global CopyId.

Entity	How one object is identified	ER choice
BOOK	ISBN alone	Regular
COPY	Its book's ISBN + CopyNo	Weak; CopyNo is partial
MEMBER	MemberId alone	Regular
LOAN	LoanId alone	Regular

Described objects	Same object?
Book B1 copy 1; book B2 copy 1	No: different owners.
Loan L1 and loan L2 for the same copy	No: different borrowing events.

B1 and B2 are short classroom aliases for ISBN values, not claims that real ISBNs have this format.

Remember: COPY is weak; LOAN is regular even though both require related objects.

Translate each rule in both directions

Write minimum and maximum before drawing the complete ERD.

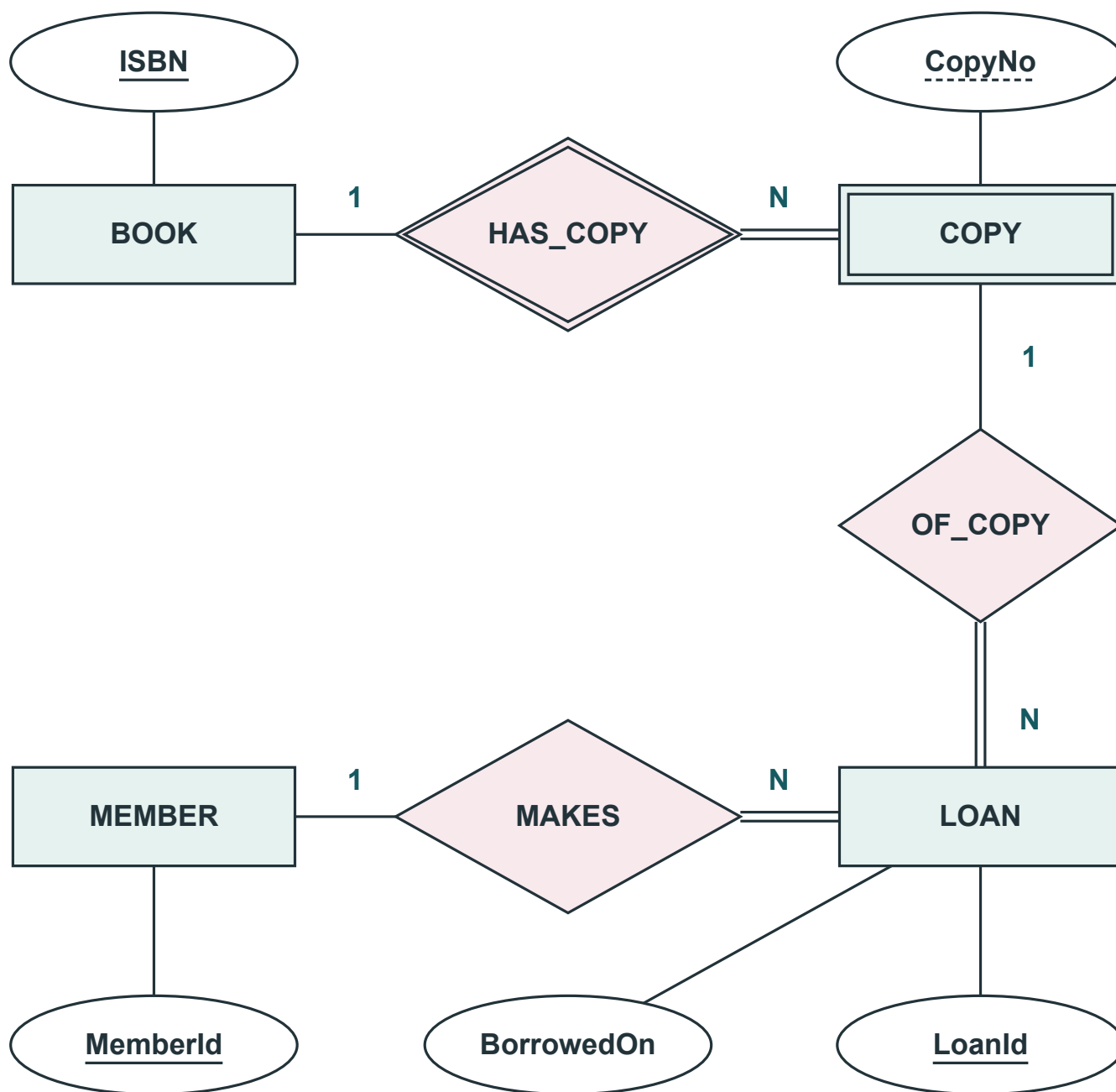
Relationship	Read one entity on the left	Read one on the right
BOOK HAS_COPY COPY	BOOK: 0 or many copies	COPY: exactly 1 book
MEMBER MAKES LOAN	MEMBER: 0 or many loans	LOAN: exactly 1 member
COPY OF_COPY LOAN	COPY: 0 or many loans	LOAN: exactly 1 copy

1. HAS_COPY supplies COPY's owner identity: use an identifying diamond.
2. MAKES and OF_COPY do not supply LOAN's identity: LoanId already does.
3. Put BorrowedOn on LOAN. Two loans made by the same member can have different dates.

Remember: Each COPY has a BOOK; each LOAN has both a MEMBER and a COPY.

The complete ERD for R1-R5

All four types, five required attributes and three relationships are shown.

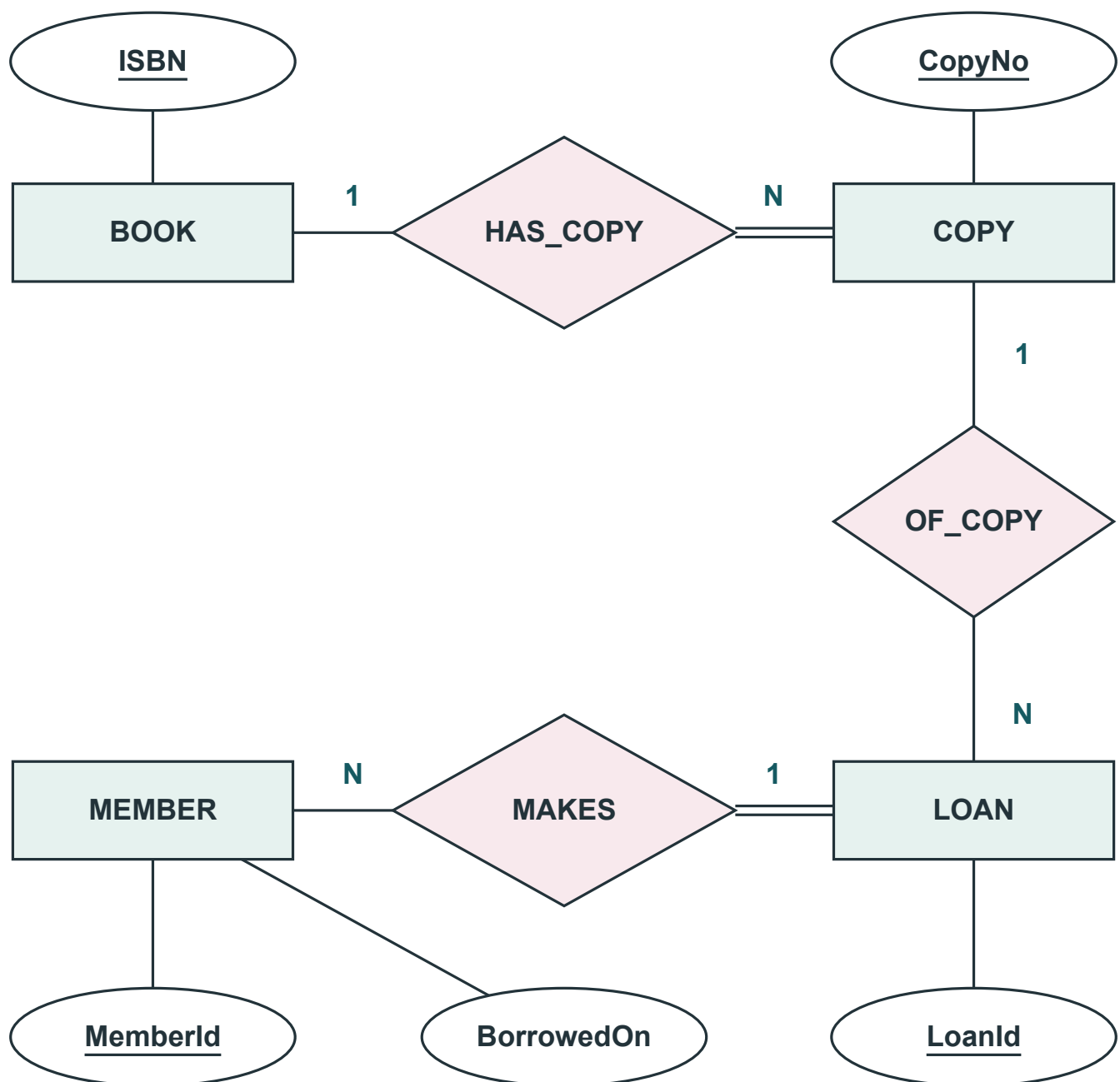


Maximum ratios and single/double participation lines. OF_COPY links a copy to its loan events.

Remember: Trace each requirement to one visible key, shape, attribute or connection.

Find four errors in this ERD

INCORRECT example. Keep R1-R5 fixed when finding the four errors.



Check identification, counts, participation and attribute placement.

Remember: For each error, name the violated requirement and a concrete counterexample.

Prove each error with a small example

Keep R1-R5 fixed. Do not change the story to make the wrong diagram work.

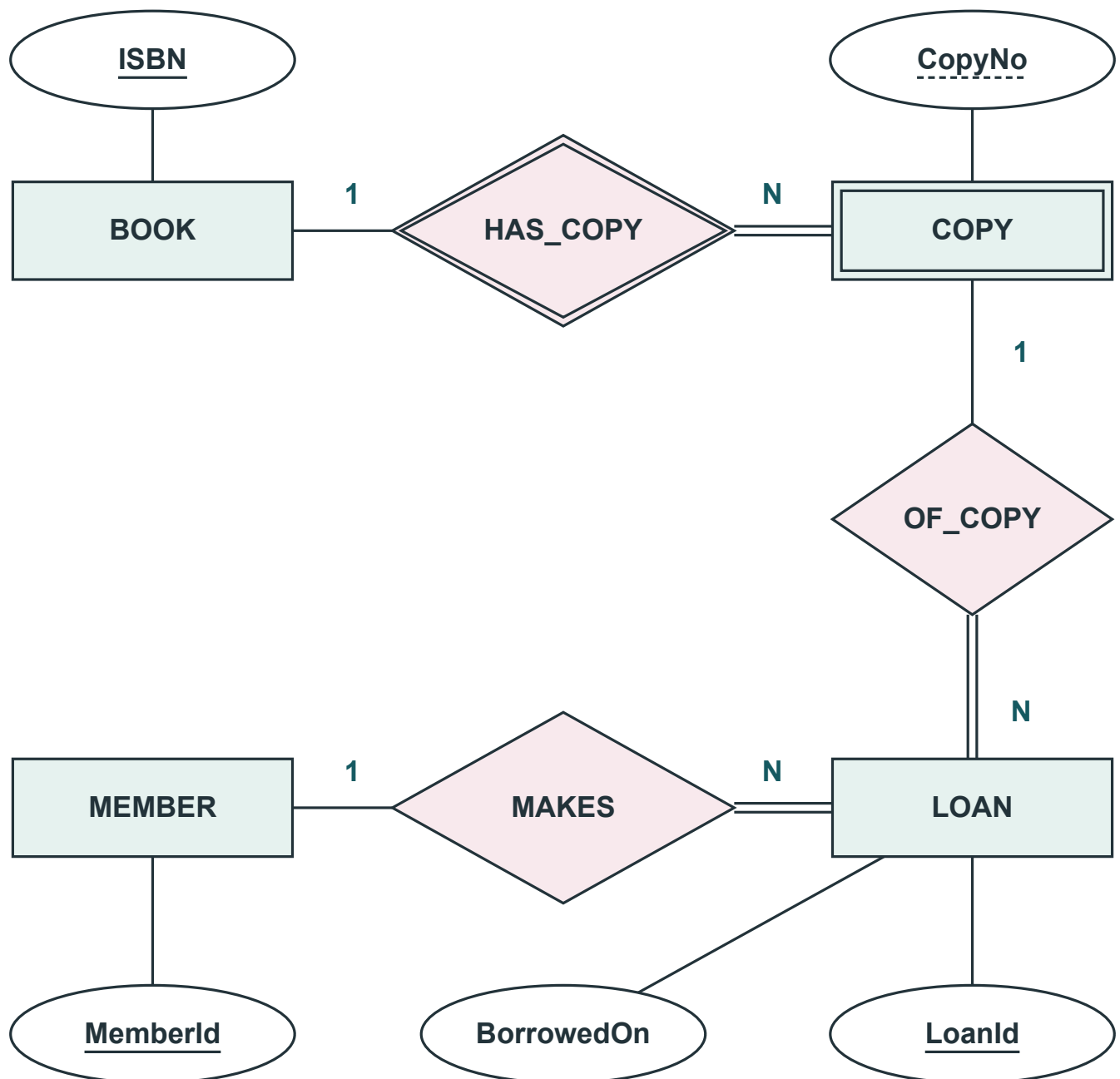
Wrong part	Counterexample	Required repair
COPY is regular; CopyNo is a full key	B1/1 and B2/1 are both allowed.	Weak COPY; partial CopyNo; identifying HAS_COPY.
MAKES ratio is N:1	One member makes L1 and L2.	1 near MEMBER; N near LOAN.
LOAN participation in OF_COPY is partial	A loan with no copy is forbidden by R4.	Double line at LOAN.
BorrowedOn is on MEMBER	M1 borrows on Sep 1 and Sep 8.	Move the date to LOAN.

BorrowedOn moving here is necessary because one member can make many loans with different dates.

Remember: A correction needs a reason, not just a differently drawn symbol.

The four repairs together

The correction returns to the complete R1-R5 design; no new ID was invented.



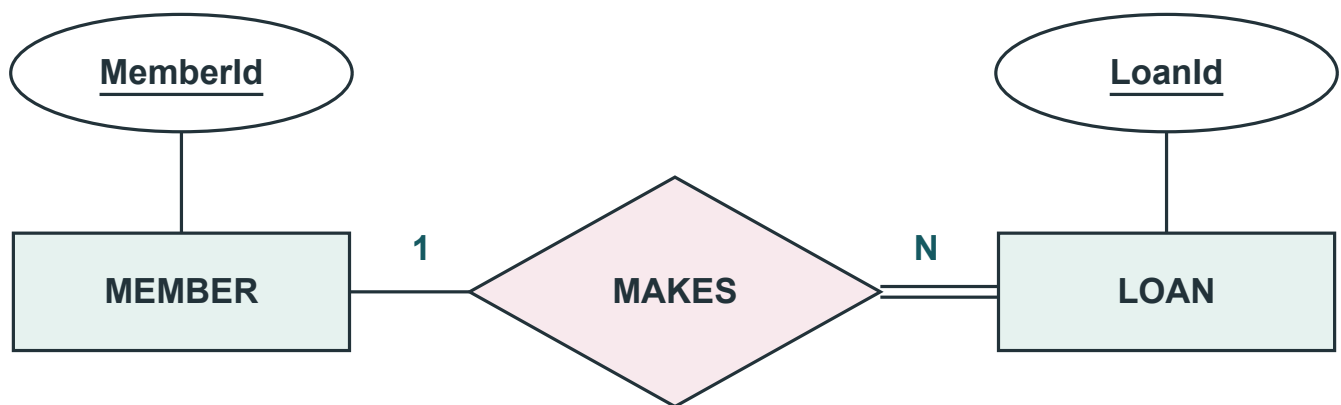
Corrected teaching example. The unchanged parts are retained so the differences are inspectable.

Remember: Compare the repaired parts against the counterexamples, one at a time.

Why not just a BORROWS diamond?

The same member may borrow the same physical copy on different dates.

LoanId	Member	Copy	BorrowedOn
L1	M1	B1 / 1	2026-09-01
L2	M1	B1 / 1	2026-09-08



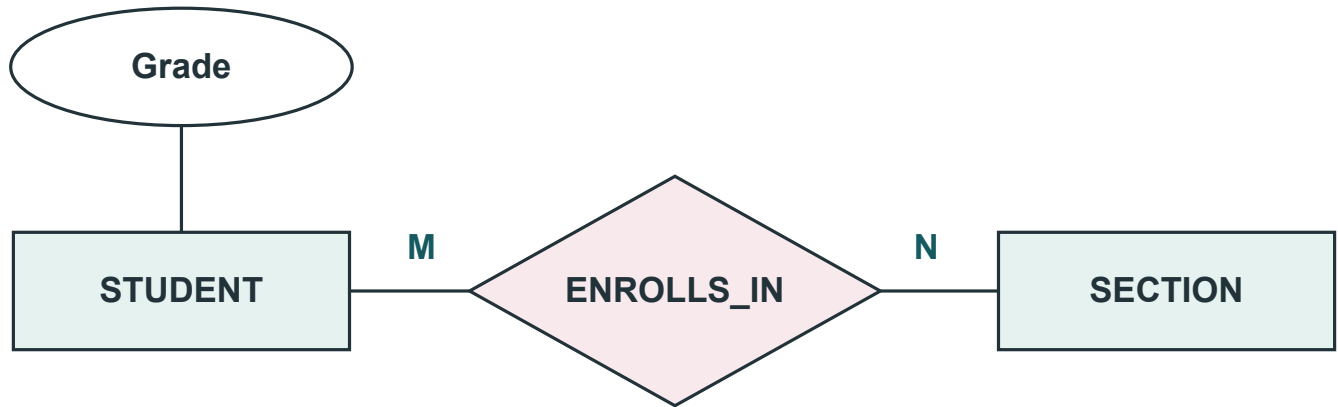
LOAN is regular with LoanId; this close-up shows only its connection to MEMBER.

1. The pair (M1, B1/1) is the same, but L1 and L2 are different events.
2. One binary relationship fact per member-copy pair cannot identify both events.
3. LOAN's unique ID preserves the distinction; neither participant needs to be duplicated.

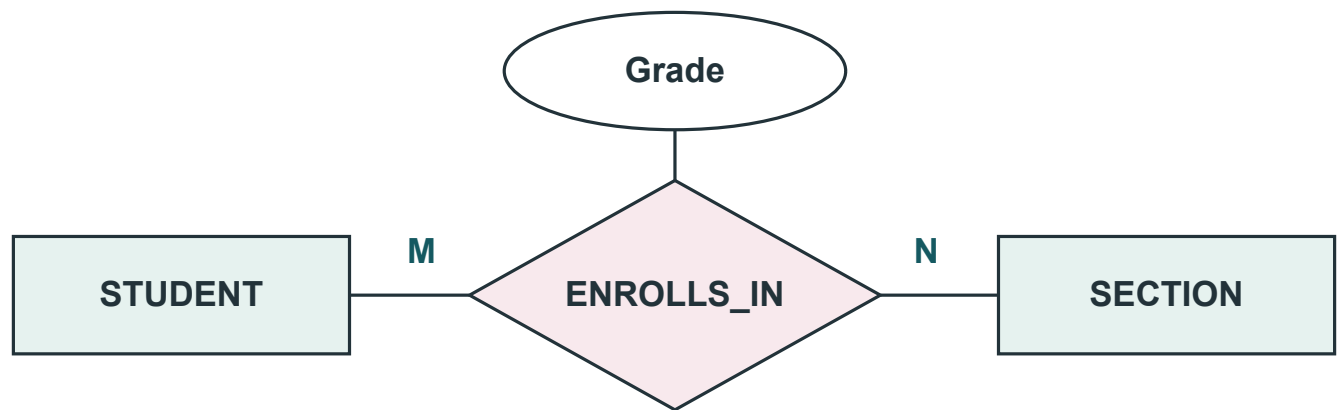
Remember: LOAN identifies events; MEMBER and COPY identify the participants.

Wrong: one Grade per student

Requirement: each enrollment has its own grade; students may enroll in several sections.



INCORRECT: Grade is attached to STUDENT. Identification is omitted in both close-ups.



CORRECT: Grade belongs to each ENROLLS_IN instance.

Student	Section	Grade
S101	DB101 / 1	80
S101	CS102 / 1	90

Remember: Use S101's 80 and 90 to prove why the upper diagram loses meaning.

Wrong: swapped min-max labels

Each section has one instructor; an instructor may teach no sections or several.



INCORRECT: this forces every instructor to teach exactly one section.



CORRECT: count the participation of the entity next to each pair.

Counterexample	Required result
I1 teaches two sections	Allowed
I3 teaches none	Allowed
A section has no instructor	Rejected

Remember: A min-max pair counts participations of the adjacent entity.

Explain this ERD in complete sentences

Read both directions. Then say what the diagram does not guarantee.



New illustrative requirement: each instructor must teach one to three sections. Keys omitted.

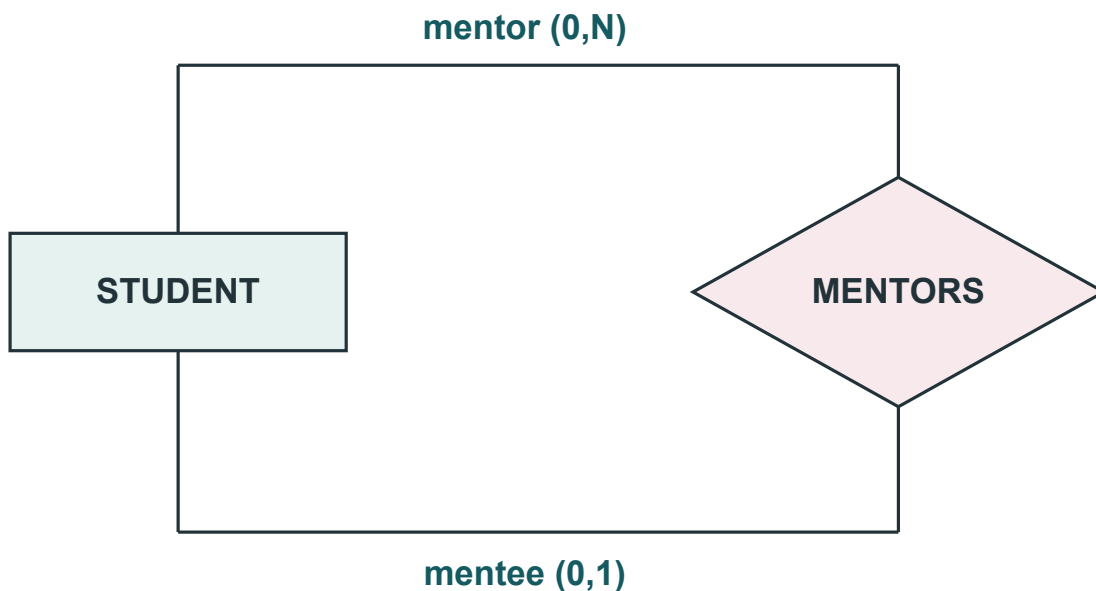
Read from	A complete explanation
One INSTRUCTOR	Each instructor teaches at least one and at most three sections.
One SECTION	Each section is taught by exactly one instructor.
Not shown	The diagram does not prevent time clashes or show the current number of sections.

Instructor counts 1 and 3 are allowed; 0 and 4 are not. A legal range is not an observed count.

Remember: Include the entity, relationship, minimum and maximum in each sentence.

Explain the two roles, not two student types

Use the role labels to say who mentors whom.



Same STUDENT type, two roles; this diagram alone does not forbid self-mentoring.

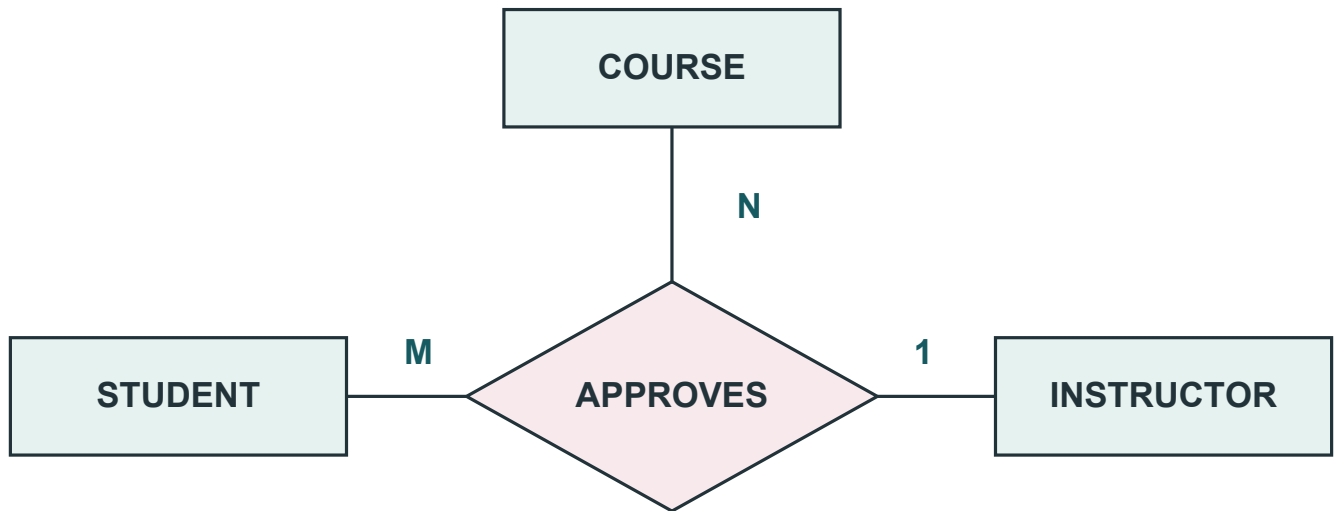
Role	Complete reading
Mentor	A student may mentor zero or many students.
Mentee	A student may have zero or one mentor.
Example	S101 mentors S102 and S103. Reversing a pair changes the fact.

Do not infer a ban on cycles or on S101 mentoring S101. Those require separate stated rules.

Remember: One entity type can play two roles in one binary relationship.

Explain the 1 without changing its meaning

This is a ternary relationship, so first name the fixed pair.



Maximum-ratio notation, not min-max. Keys are omitted.

Statement	Supported?
For one student and one course, at most one instructor approves.	Yes
I1 can appear in only one approval overall.	No
Every student must receive an approval.	No: total participation is not required here.

I1 may approve S101 for DB101 and S102 for CS102: the fixed student-course pairs differ.

Remember: One instructor per student-course pair is not one approval per instructor.

Three ways to demonstrate ERD understanding

The same concepts support drawing, correcting and explaining.

Task	What a complete response shows
Description to ERD	Required types and attributes; keys; relationship names; both minimum and maximum; weak-entity identification.
Description plus wrong ERD	The wrong part; the violated requirement; a small counterexample; a corrected diagram.
ERD to explanation	Both directions; roles; identity; optional/required participation; what is not implied.

1. Do not invent restrictions missing from the description.
2. Do not silently replace the stated model with a different model.
3. For each correction, explain why the repaired drawing now fits the requirement.

Remember: Always connect your answer to the given requirements or visible symbols.

Test the drawing with these four states

Return to the small Fall 2026 campus model, not UNIVERSITY's minimum of five.

Proposed state	Expected judgment	Reason
DB101 / 1 and CS102 / 1	Allowed	Different owners
I3 with no teaching assignment	Allowed	INSTRUCTOR participation is partial
A section with two instructors	Not allowed	Maximum one instructor
S103 with no enrollment	Allowed	Enrollment is optional

1. Each group draws the same required types and relationships, then explains its identification choice.
2. After responses are fixed, compare all anonymous responses individually using these checks.
3. After instructor feedback, revise your drawing. Rankings do not determine grades.

Keep your drawing, one prediction, the observed result, and a correction or explanation. Submit only assigned work.

Remember: A drawing is useful when you can explain which states its rules permit.

Change one rule, then find the changed symbol

Keep every other requirement fixed.

Change	What to inspect
Allow several instructors per section	Maximum instructor count per section
Require every student to enroll	Minimum STUDENT participation in ENROLLS_IN
Keep sections from several terms	Whether course + section number still identifies one section
Give every section a global SectionId	Whether identification still needs an owner
An instructor must teach 1-3 sections	(1,3): counts 1 and 3 pass; 0 and 4 fail

A new global ID does not remove required course ownership. A team-teaching rule does not automatically change student enrollment.

Remember: Explain the rule before changing the diagram.

What should you remember from Ch3?

Use these questions to explain a diagram without reading long paragraphs.

Ask	Point to or explain
What object and what type?	One instance versus the type rectangle
What identifies it?	Complete key, or owner plus partial key
What is connected?	Relationship and named roles
How many may or must connect?	Maximum and minimum, separately
Whose property is this?	Entity attribute or relationship attribute
Which rules changed?	Small campus versus UNIVERSITY

1. Use the same small data when comparing two interpretations.
2. For a ternary constraint, distinguish a fixed pair from one entity's total participation.
3. SQL and relational mapping come later; neither is needed to explain these ER drawings.

Remember: Ch4 extends ER with shared properties and more specialized entity types.

Textbook sections behind these examples

Elmasri and Navathe, Fundamentals of Database Systems, 7th Edition, Pearson.

Topic	Sections	Photographed pages
Requirements and conceptual design	3.1-3.2	90-93
Entities, attributes, keys, domains	3.3	93-102
Relationships, roles, constraints	3.4	102-108
Weak entities; refinement; ER notation	3.5-3.7	109-115
Ternary relationships and constraints	3.9.1-3.9.2	118-122
UNIVERSITY example; Figure 3.20	3.10	122-124

1. Original English explanations, diagrams and illustrative data adapt the existing course examples.
2. Section 3.8 UML is not assigned in this chapter.
3. The UNIVERSITY prose/figure difference is retained explicitly. This teaching PDF is not a publisher's slide deck.

Remember: Use section numbers when another printing has different page numbers.